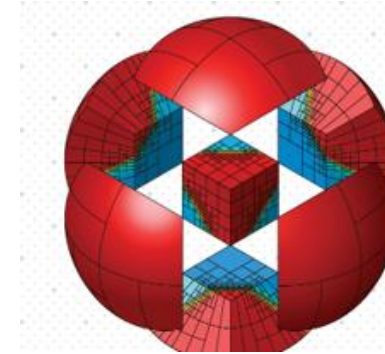




Scalable phase-field fracture simulations on Frontier using MFEM

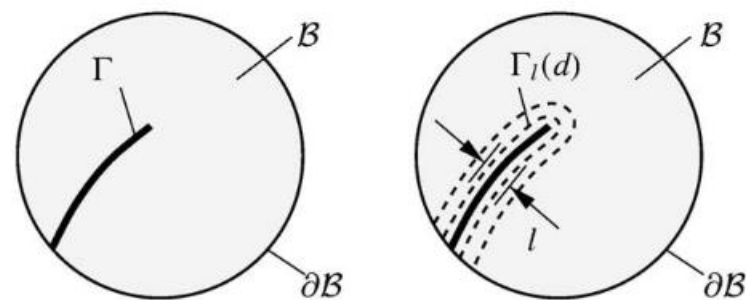
Ashvin Oli, Aussie Greene, Abhinav Gupta, Ravindra Duddu

Civil Engineering
Vanderbilt University, Nashville, TN

MOTIVATION

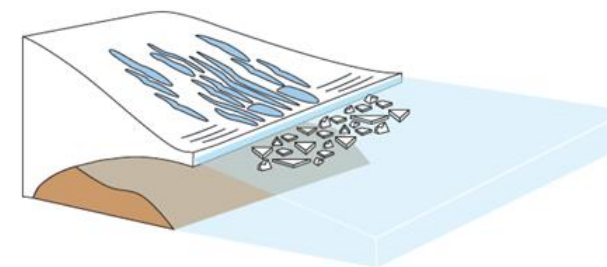
Exascale fracture in glaciers

- CPML Lab
- Work primarily in fracture mechanics
- Use Phase field fracture
- Application in glacier fracture

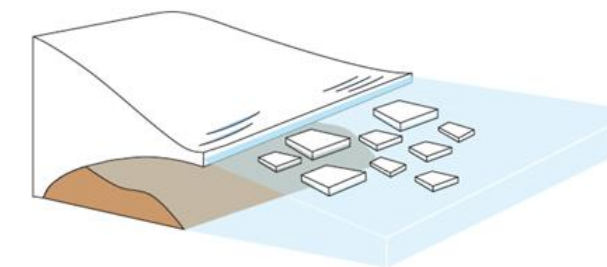


Ice shelves: floating calving events

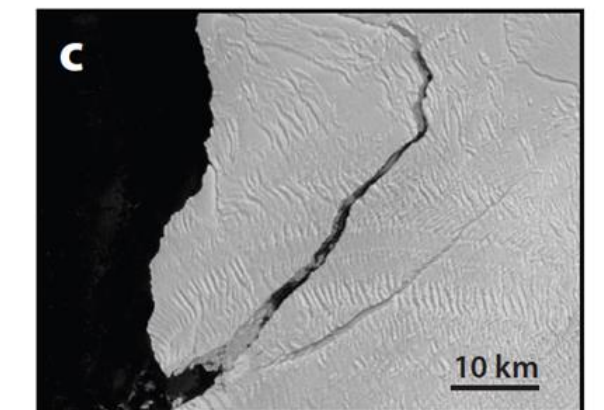
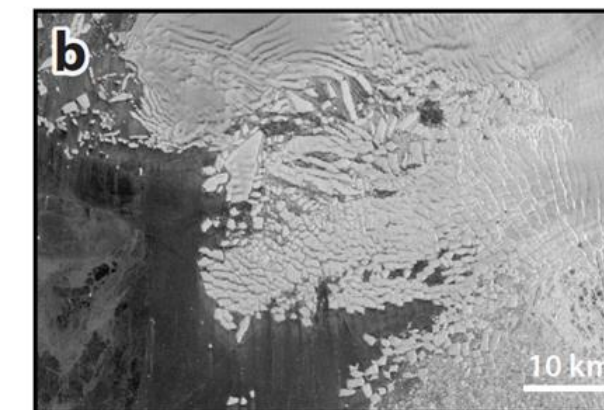
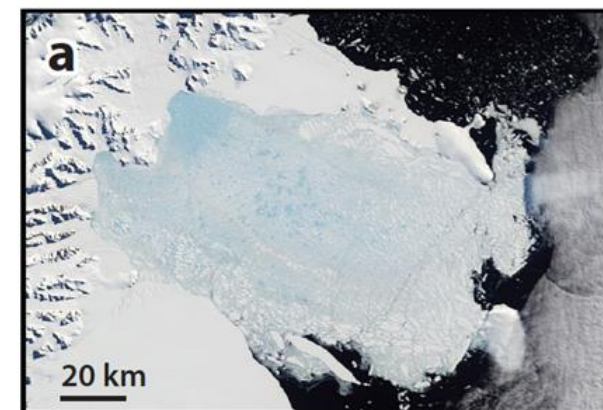
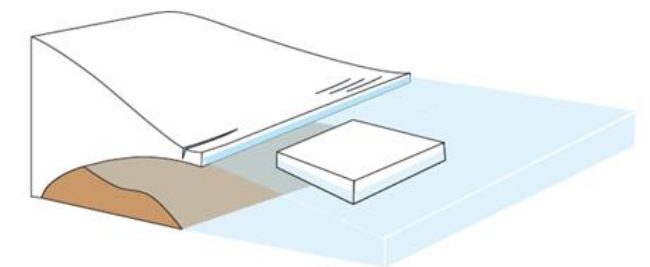
Explosive disintegration
Larsen B Ice Shelf



Sustained calving of icebergs
Thwaites Glacier



Rifting of tabular icebergs
Brunt Ice Shelf



WEEKS

DECADES

Bassis et al, 2024

A phase-field crack must be resolved with elements $h \leq \frac{\ell}{4}$;
One km-scale domain (with uniform refinement) is of billions
of unknowns.

Governing equations

Elasticity — find u

$$\nabla \cdot \boldsymbol{\sigma}(u, d) + \mathbf{b} = 0$$

$$\boldsymbol{\sigma} = g(d) \mathbf{C} : \boldsymbol{\varepsilon}(u)$$

$$g(d) = (1 - d)^2 + \eta$$

Dirichlet displacement on $\partial\Omega_D$, traction on $\partial\Omega_N$. Small strain, isotropic degradation of the full elastic stress.

Damage — find $d \in [0, 1]$

$$\ell^2 \nabla^2 d - d + (1 - d) \mathcal{H}^+ = 0$$

$$\nabla d \cdot \mathbf{n} = 0 \text{ on } \partial\Omega$$

$$\mathcal{H}^+ = \max(0, \Sigma \langle \sigma_i^{\text{eff}} \rangle_+^2 / \sigma_c^2 - 1)$$

Stress-threshold driving force, made non-decreasing for irreversibility. Length scale ℓ regularizes the crack.

Two elliptic, symmetric positive definite problems

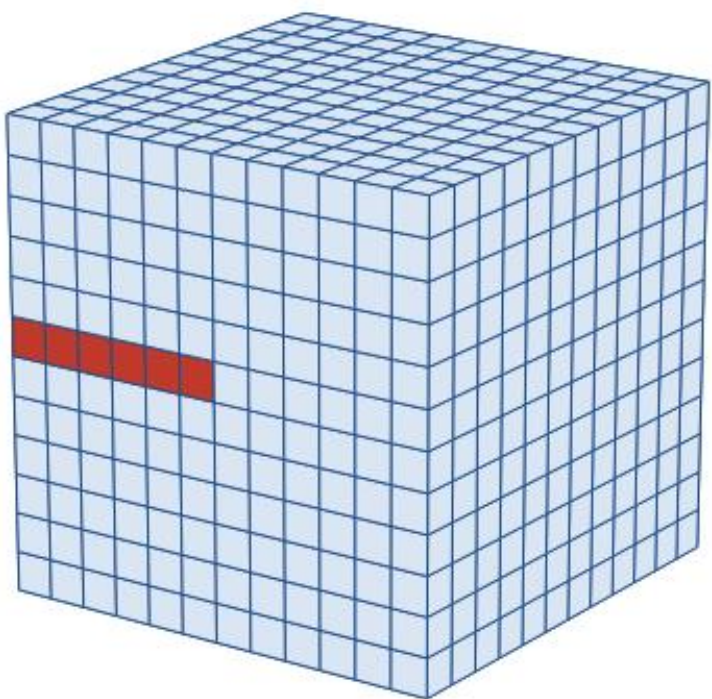
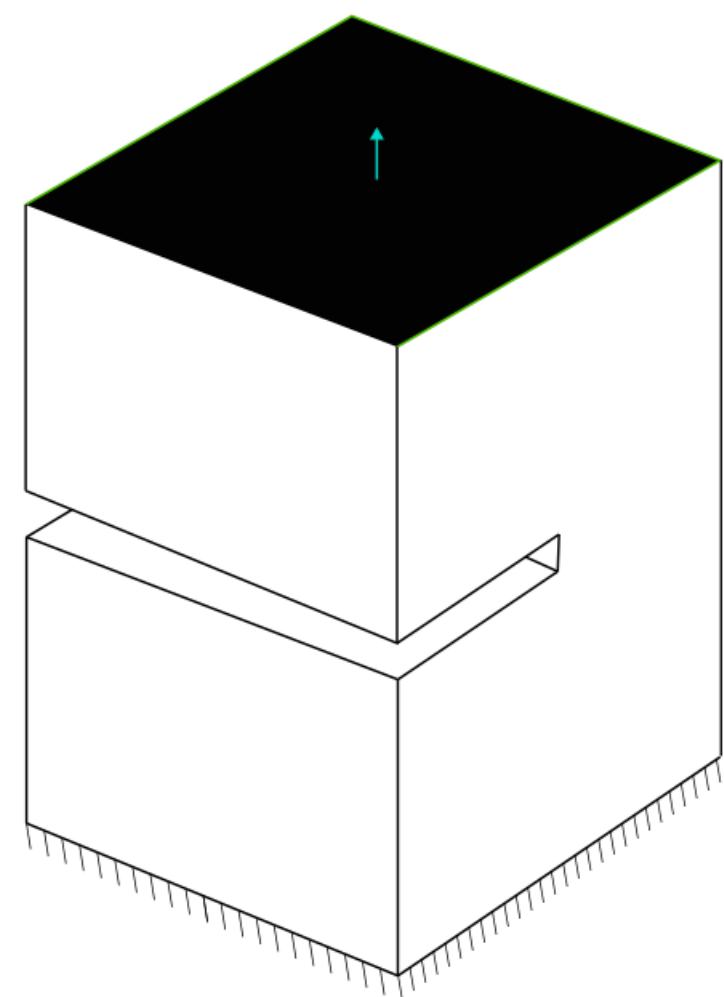
$$a(u, v) = \int \boldsymbol{\varepsilon}(u) : g(d) \mathbf{C} : \boldsymbol{\varepsilon}(v)$$

Symmetric: $g(d) \mathbf{C}$ inherits the major symmetry of $\mathbf{C}$. Positive definite: $g(d) > 0$ everywhere, by the η residual floor.

$$\tilde{a}(d, \omega) = \int (1 + \mathcal{H}^+) d \omega + \ell^2 \int \nabla d \cdot \nabla \omega$$

Symmetric by inspection. Positive definite because $\ell > 0$ and $(1 + \mathcal{H}^+) \geq 0$ — a reaction–diffusion operator.

Problem Geometry and Boundary conditions



Material Properties

Young's modulus, E	210
Poisson's ratio, ν	0.3
σ_c	4
ℓ	0.035
Load increment, Δu	1×10^{-5}
Total load increments	5

Three Base Meshes, Refined Twice

Chain A → 239^3	NP 40 13.6M (41.5M / 13.8M)	NP 320 109.0M (329.4M / 109.8M)	NP 2560 871.9M (2.63B / 875.1M)
Chain B → 302^3	NP 80 27.5M (83.5M / 27.8M)	NP 640 220.0M (663.8M / 221.3M)	NP 5120 1.76B (5.29B / 1.76B)
Chain C → 380^3	NP 160 54.8M (165.9M / 55.3M)	NP 1280 438.4M (1.32B / 440.4M)	NP 10240 3.51B (10.55B / 3.52B)

- 3 base meshes ($1 \times 1 \times 1$) (Hex mesh) which are refined progressively during the runtime to get the required number of DoFs/Elements

From weak form to MFEM

$$\int \varepsilon(u) : g(d)\mathbf{C} : \varepsilon(v) = \int \mathbf{b} \cdot v + \int_{\partial\Omega_N} \mathbf{t} \cdot v$$

```
ParBilinearForm a(&fespace);
a.AddDomainIntegrator(
    new ElasticityIntegrator(lambda_d, mu_d));
```

λ, μ enter as coefficients already scaled by $g(d)$ — damage coupling needs no new integrator.

$$\int (1+\mathcal{H}^+)d\omega + \ell^2 \int \nabla d \cdot \nabla \omega = \int \mathcal{H}^+ \omega$$

```
a.AddDomainIntegrator(new MassIntegrator(rho_coef));
a.AddDomainIntegrator(new DiffusionIntegrator(lc2_coef));
a.AddDomainIntegrator(new MassIntegrator(H));
```

PETSc/GAMG over hypre/BoomerAMG

Same 1.4M-element mesh, matched Krylov settings.

Ranks	Backend	Setup	ELASTIC (S)			PHASE (S)	
			Asm	FormLinSys	Solve	Asm	Solve
8	PETSc/GAMG	5.967	9.555	9.102	13.760	3.798	1.940
	HYPRE/BoomerAMG	5.728	8.915	22.424	85.346	4.315	0.413
64	PETSc/GAMG	6.483	1.179	1.236	2.422	0.383	0.334
	HYPRE/BoomerAMG	5.846	1.079	1.521	12.884	0.461	0.077
256	PETSc/GAMG	7.142	0.294	0.319	1.300	0.091	0.128
	HYPRE/BoomerAMG	6.767	0.266	0.250	3.886	0.100	0.040

3.0–6.2× on the elastic solve, 2.1–2.9× on total time, across a 32× rank range. Iterations are not the cause — 39–44 vs. 31–33.

- We came from FEniCS , already on PETSc — options files, per-operator prefixes and solver vocabulary carried over
- Flags are runtime, not compile time. Swapping a preconditioner needs no rebuild

* How times are reported: Using MPI_Wtime() calls and MPI_reduce using MPI_MAX

CG + GAMG for both operators

Solver flags used

Both Krylov solves	cg + gamg	
Aggregation	agg (smoothed)	
Level smoother	chebyshev / jacobi	
Coarse solve	preonly / telescope	
Tolerance	rtol 1e-7	
Elastic max iterations	20 (capped)	Chafia et al, 2024
Per-operator prefixes	-elastic_ / -phase_	

SPD operators, so CG is the right default — coarse_eq_limit and threshold are left unset, so PETSc's own compiled GAMG defaults (≈ 50 , -1) pick the coarsening depth automatically.

PLATFORM

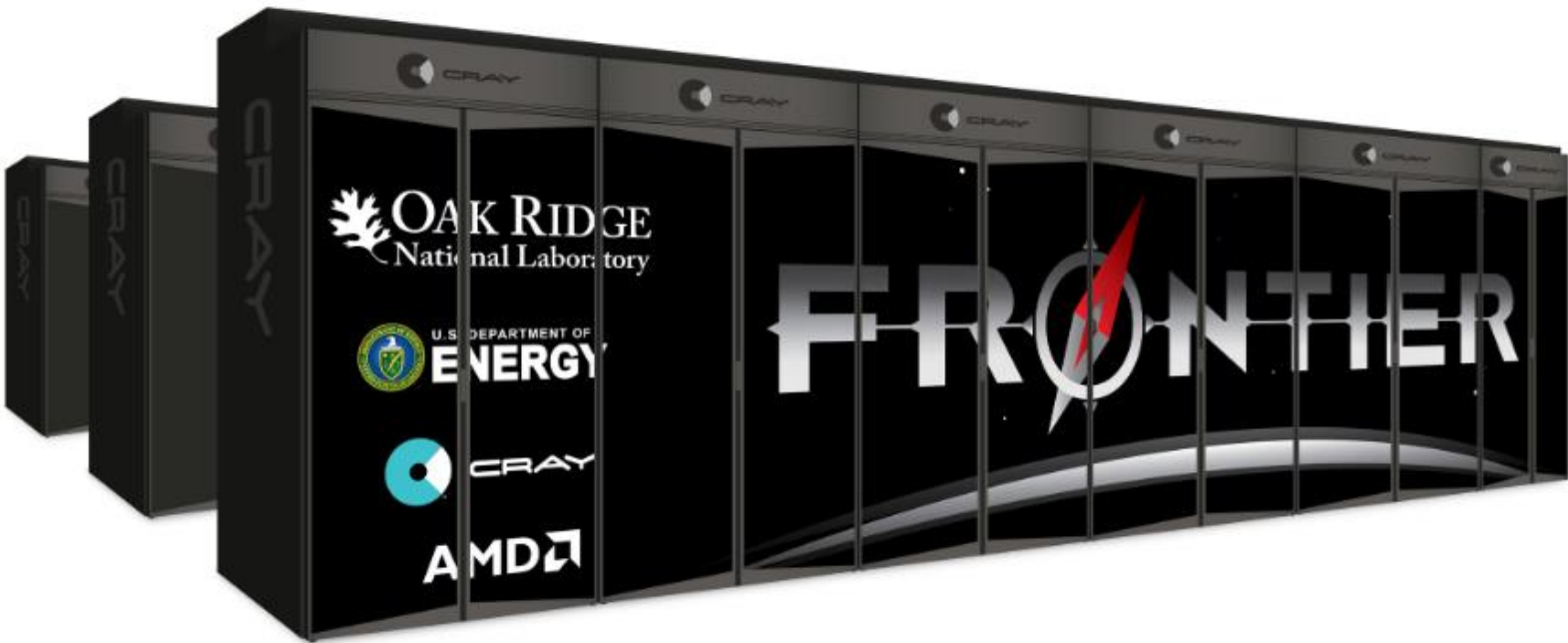
The machine these runs sit on



Nodes	9,856 · HPE Cray EX235a
Per node	64-core AMD EPYC + 4 × MI250X
Memory	512 GB DDR4 · 64 GB HBM2E per GCD
Network	4 × 200-Gbps Slingshot NICs · 100 GB/s/node
Peak	≈ 2 EF/s FP64

What we actually use

- CPU path only. MPI-everywhere, 56 allocatable cores per node



Eight ranks per node

Production mesh (232.99M elements, 702.9M displacement DOFs), **rank count pinned at 640** — only the packing changes.

Ranks/node	Nodes	Setup (s)	Elastic solve (s)	T_{total} (s)	Node- hours	Outcome
4	160	21.72	8.325	1670.5	75.2	Completed
8 ◀	80	21.19	8.665	1671.7	37.6	Completed
16	40	25.24	11.948	2103.4	23.7	Completed
32	20	—	—	—	—	Out-of-memory
56	12	—	—	—	—	Out-of-memory

Baseline profile, $n = 1$ per density. “Elastic solve” is one solve; T_{total} is the whole job — different units, not directly comparable.

4 and 8 are **tied on wall clock** (1670.5 vs. 1671.7 s, <0.1% apart) — so 8 wins outright: **half the nodes, half the node-hours** .

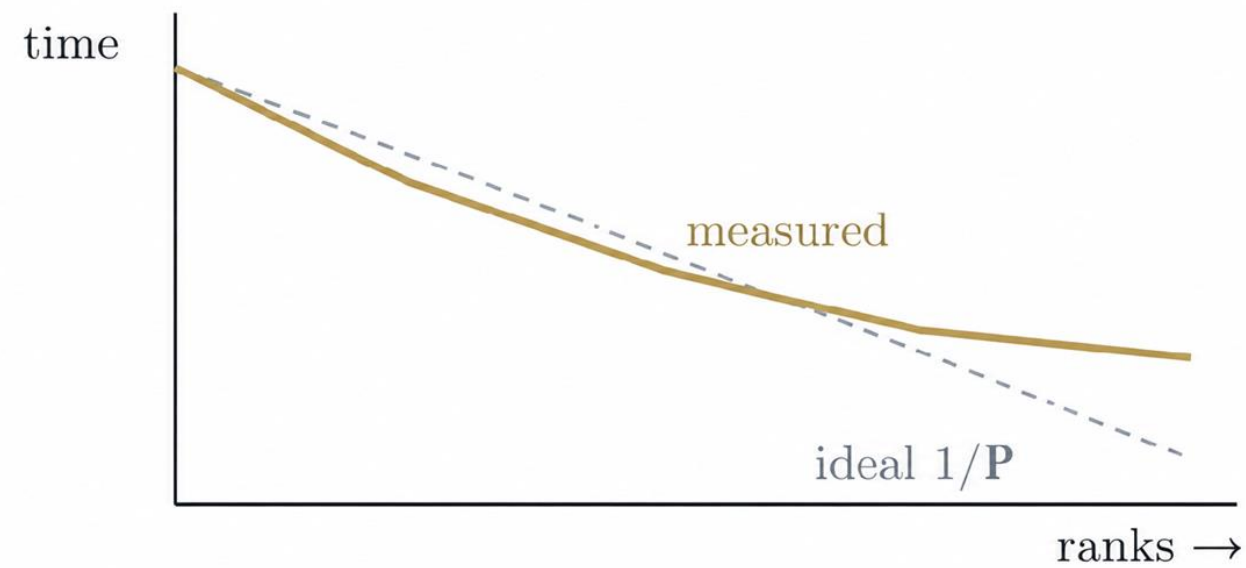
32 and 56 ranks/node were **killed by the OOM manager**.

Every number after this slide is at **8 ranks/node**.

DEFINITIONS

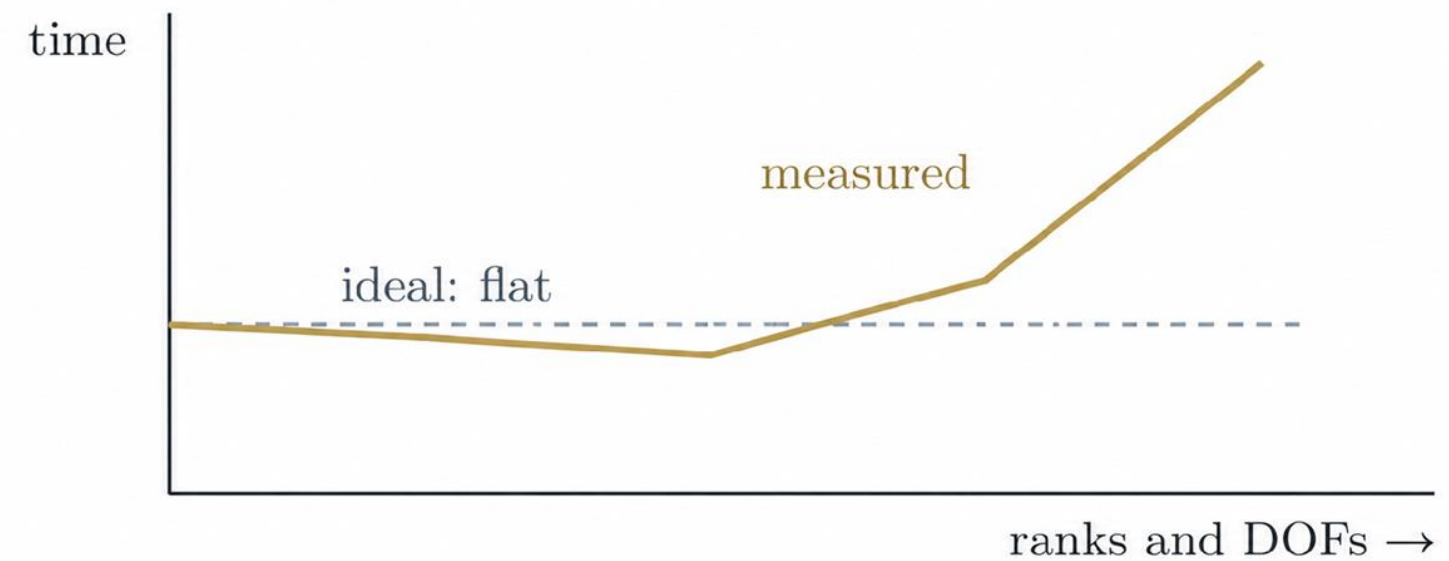
Strong and Weak scaling

Strong scaling



Fixed problem, more ranks — ideal is $\text{time} \propto 1/P$.

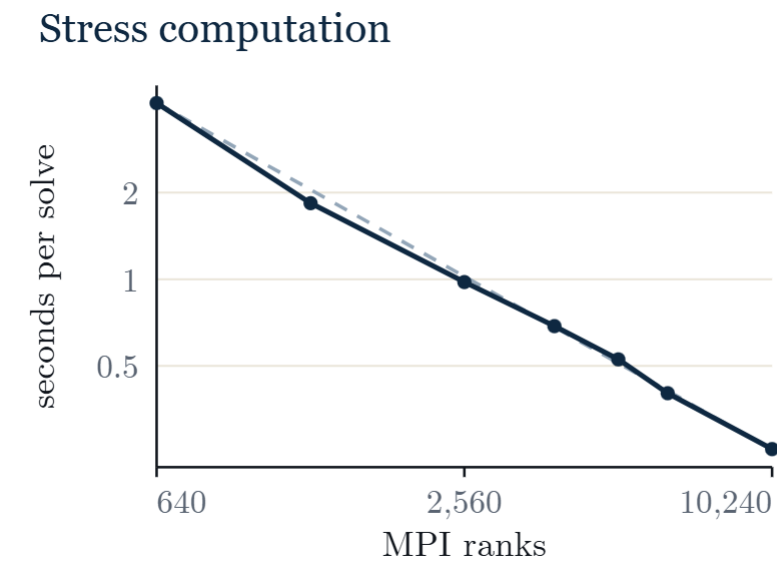
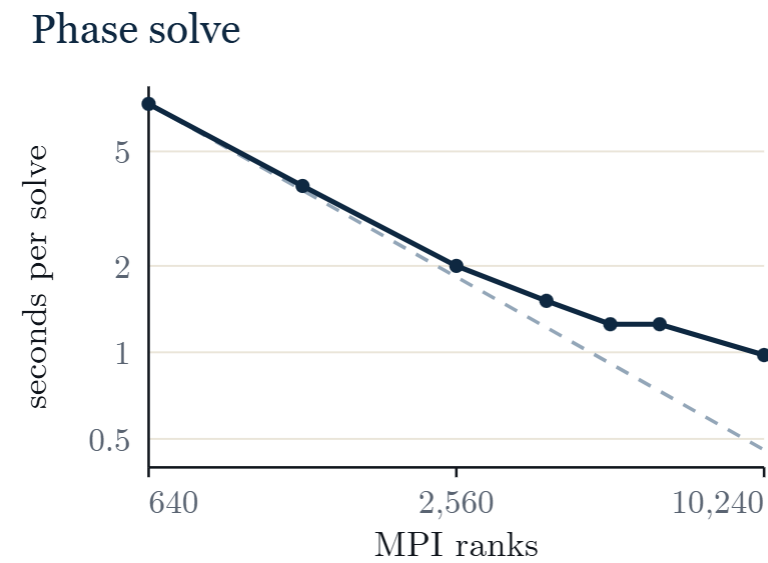
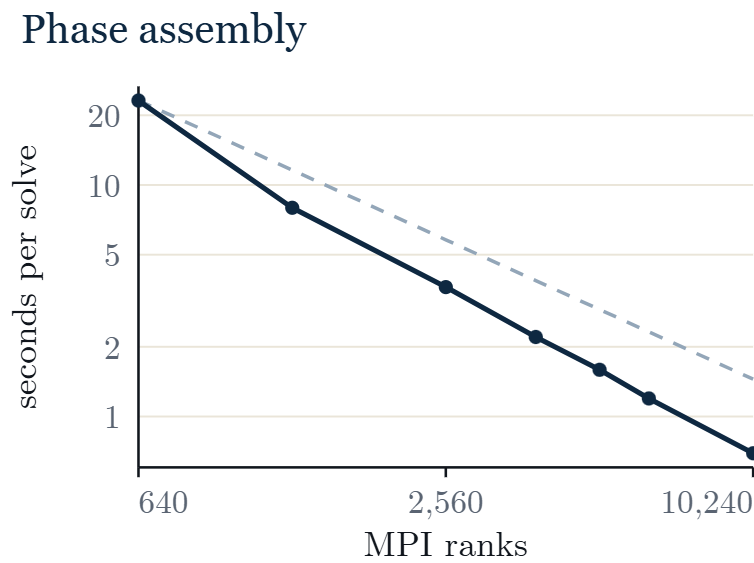
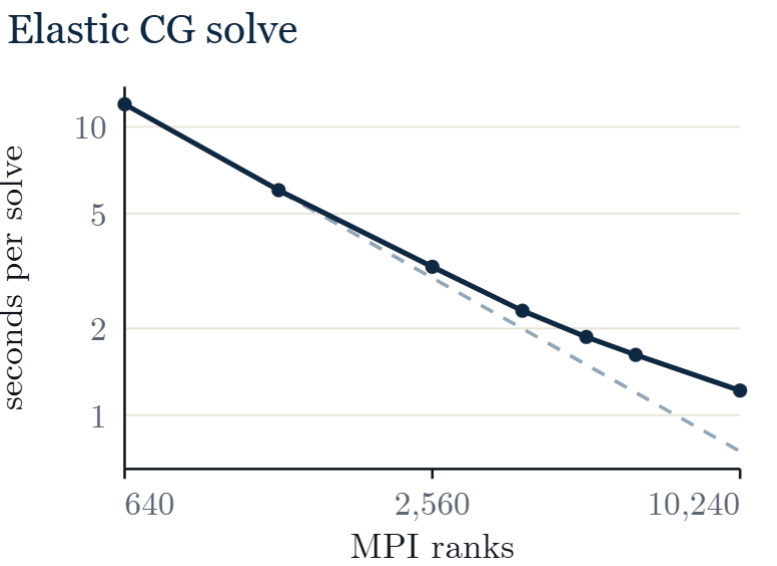
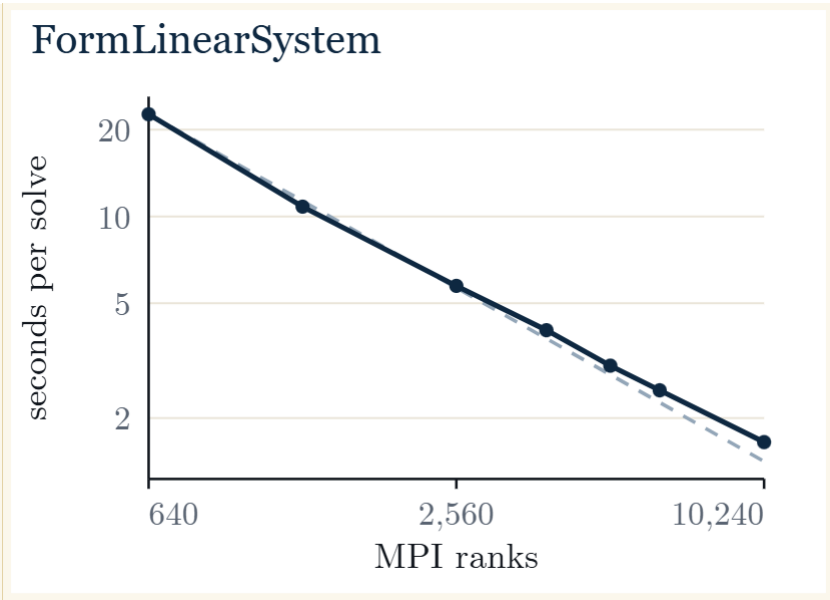
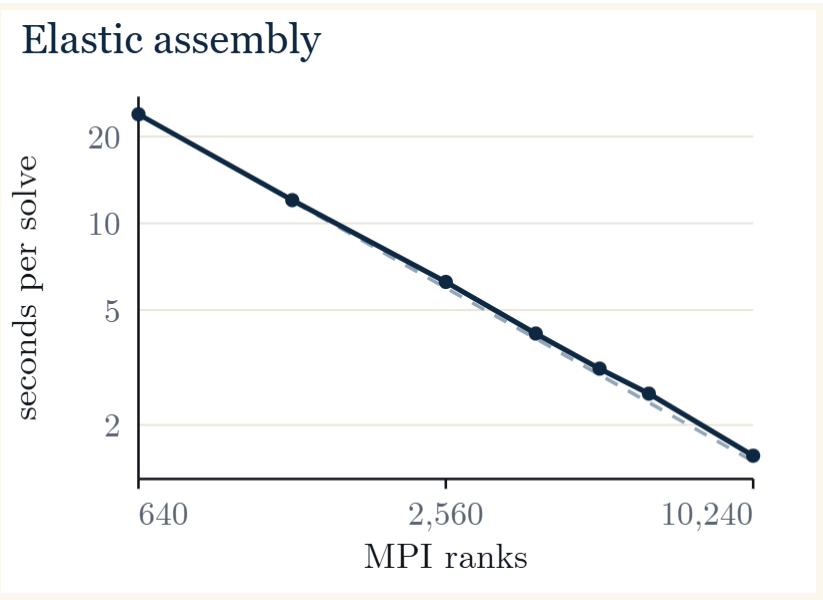
Weak scaling



Fixed work per rank, problem grows with P — ideal is flat.

Scaling is function of both the software and hardware

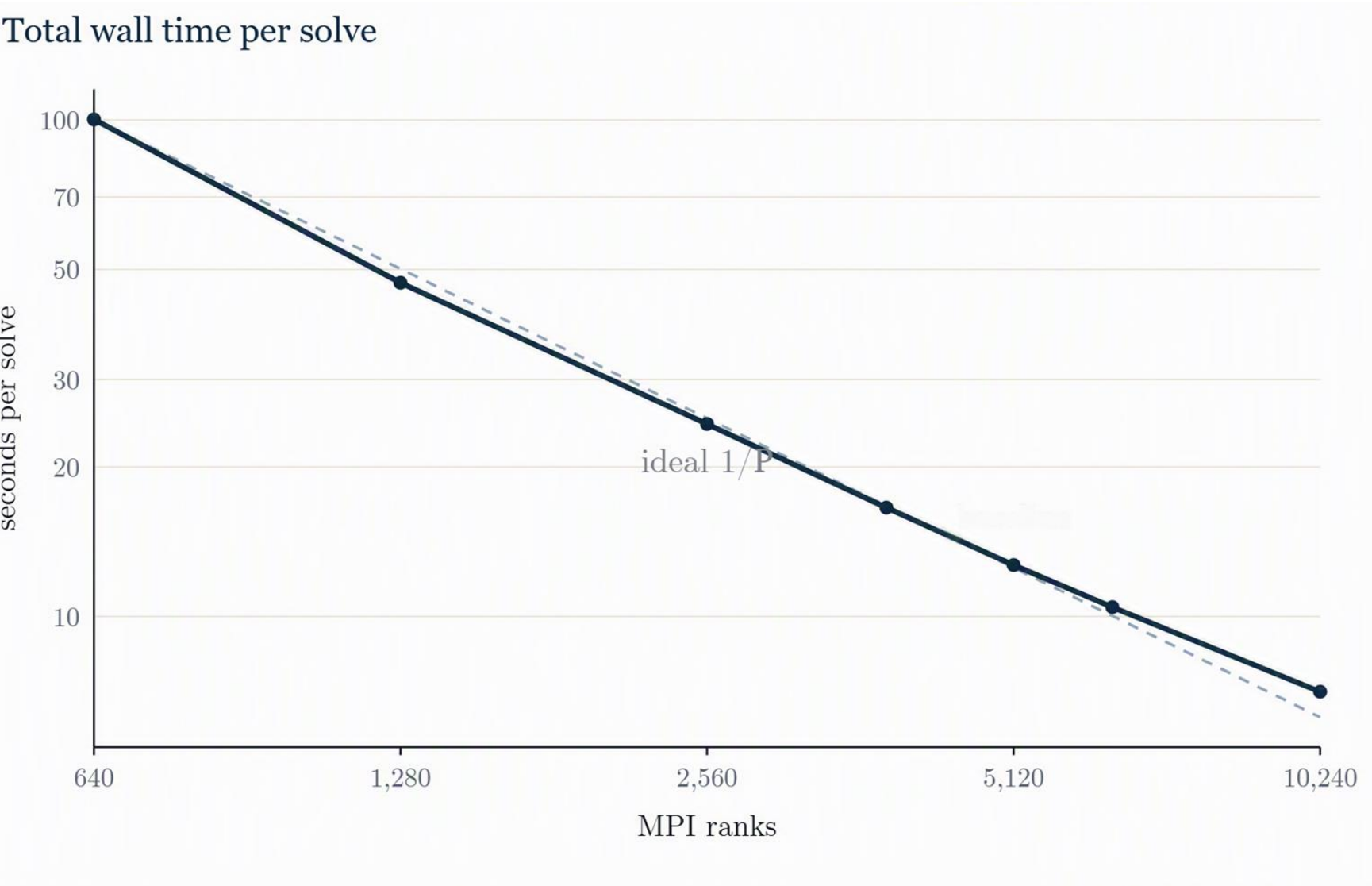
Per solve times, at fixed problem size



— **Our run** — capped at 20 CG iters; the profile shown in every cost panel **ideal** — $1/P$ from the baseline 640-rank value

Cost panels log-log, *per solve*; iteration panel log-linear. 232.99M elements fixed, 640–10,240 ranks.

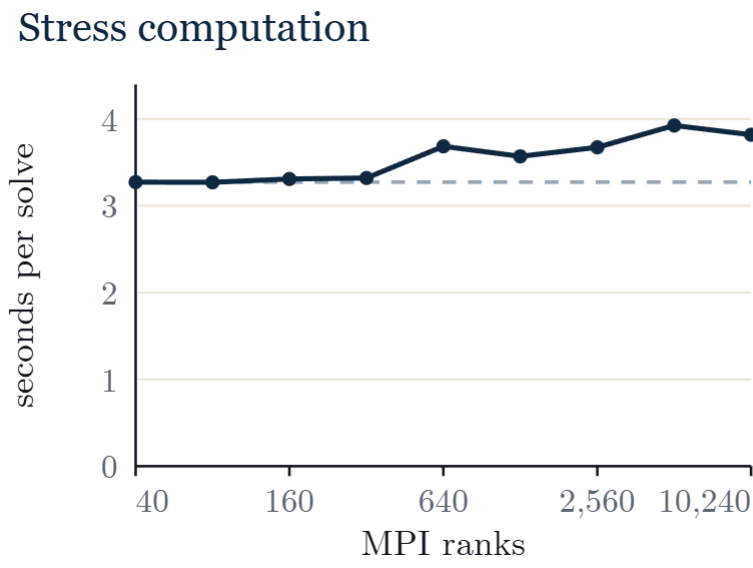
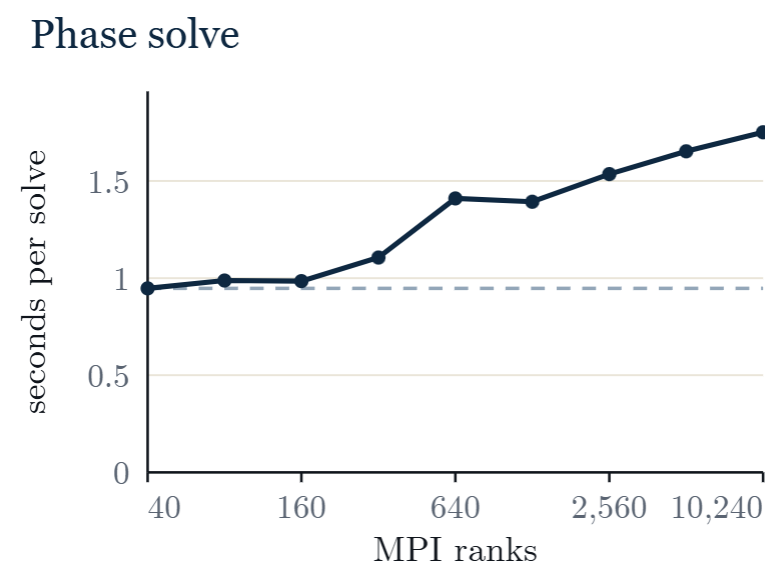
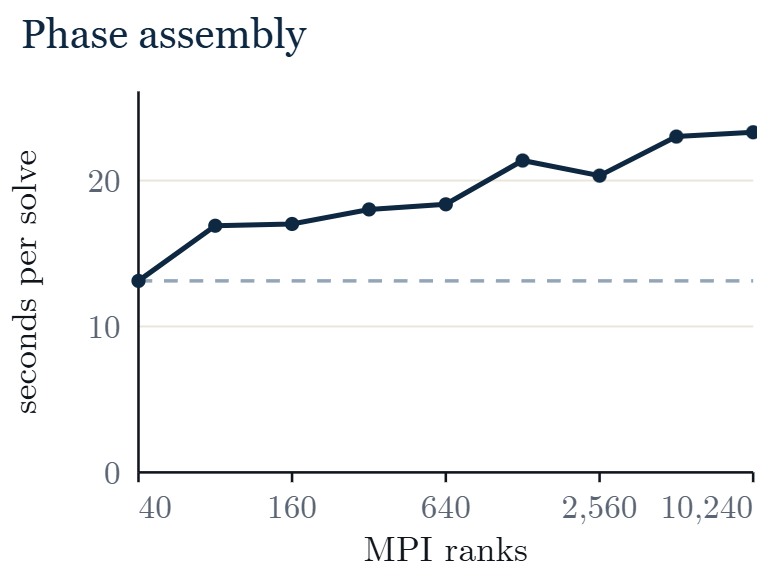
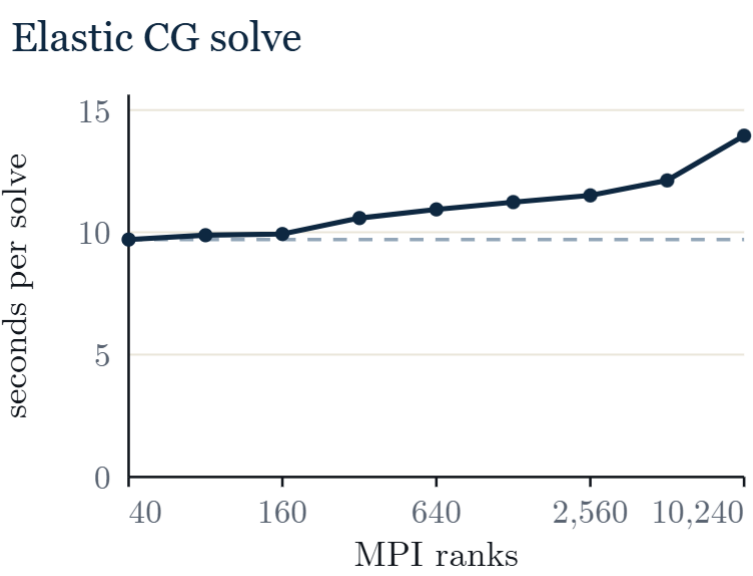
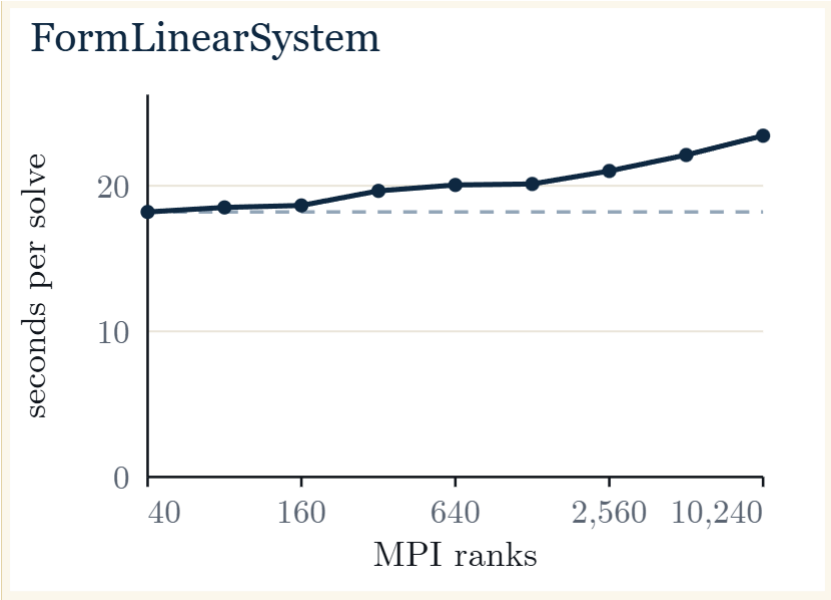
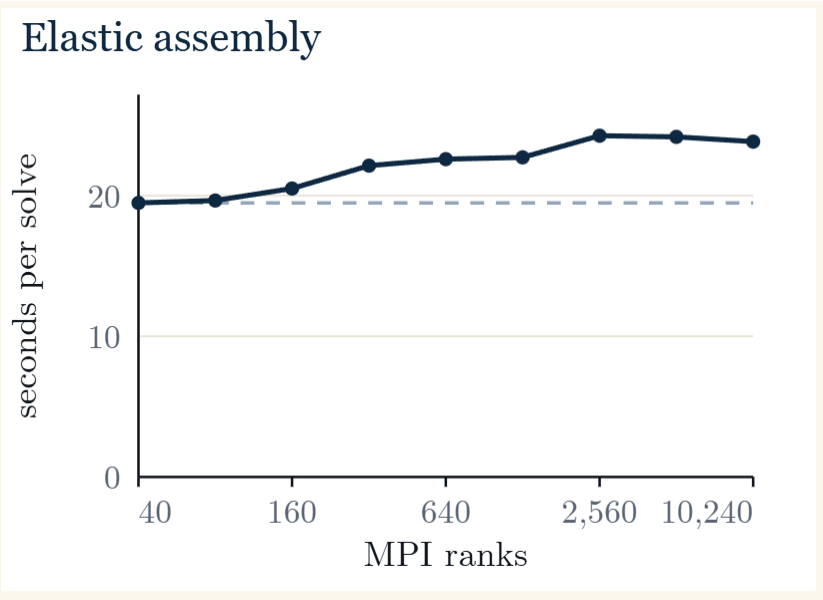
Total time per solve



Our run holds close to ideal across the whole ladder: at **10,240 ranks**, 7.05 s/solve against an ideal 6.27 s — **89% efficiency over a 16× rank increase** .

$$E_s(p) = \frac{T(640)}{T(p)} \times \frac{640}{p}$$

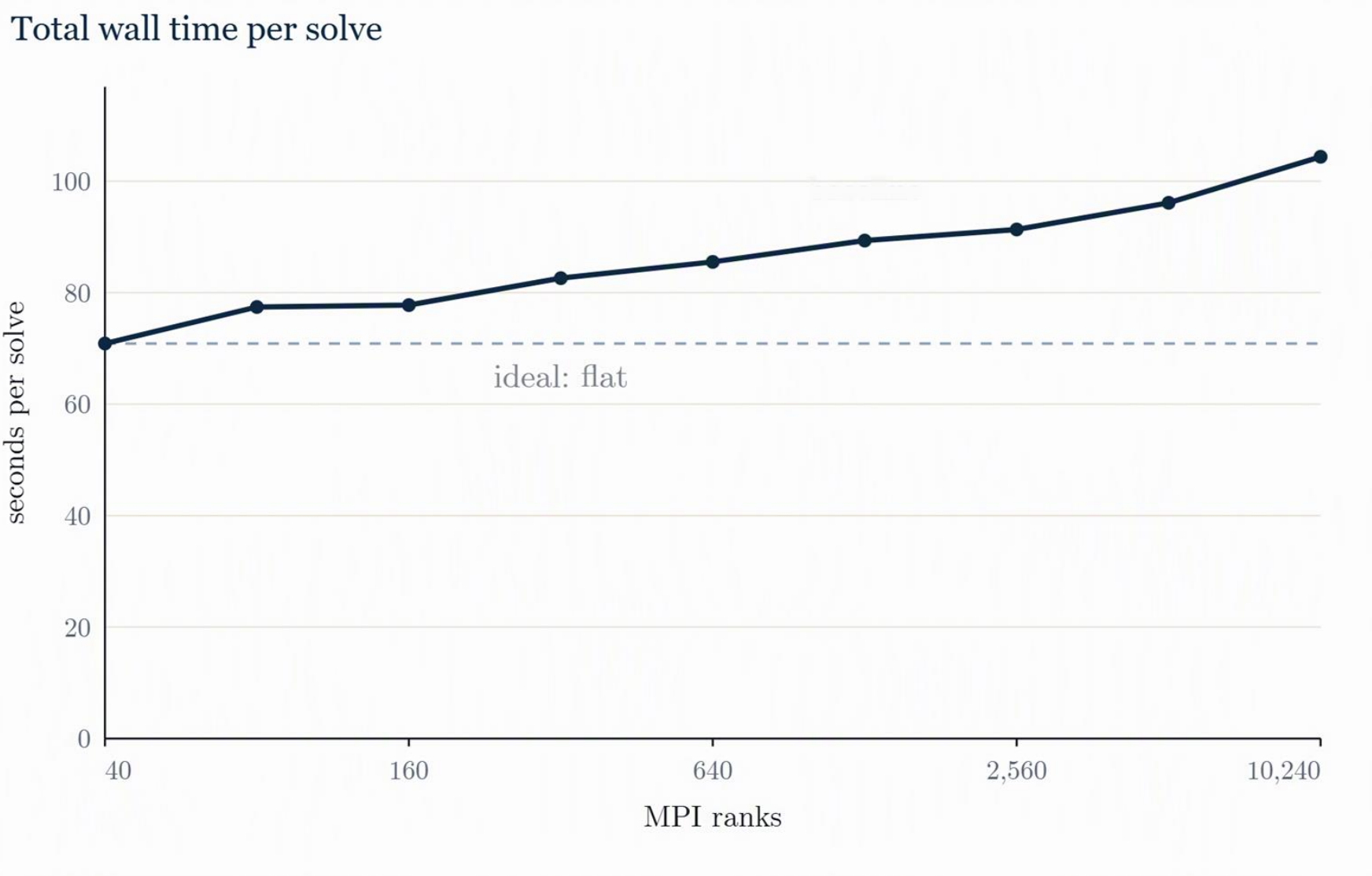
Per-solve cost at fixed work per rank



— **Our run** — capped at 20 CG iters; the profile shown in every cost panel **ideal** — flat at the baseline 40-rank value

Linear vertical axes, *per solve*; log rank axis. $w \approx 70$ elements/rank, 40–10,240 ranks (41.5M–10.55B dofs).

Total time per solve — and what sets the floor



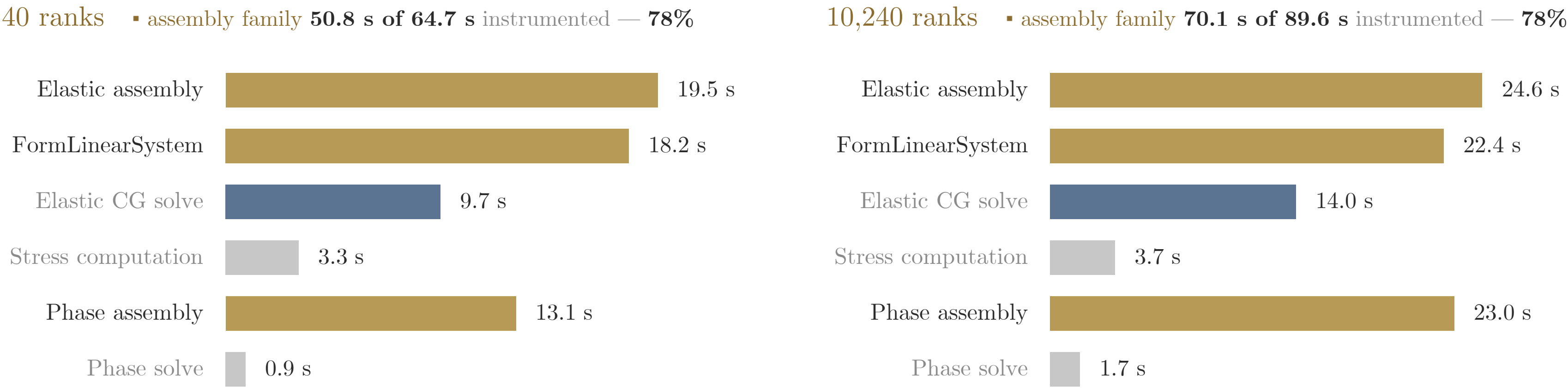
Across a **256×** growth in problem size (41.5M → 10.55B dofs), baseline ends at **1.47× ideal** — 104.4 s/solve against 70.9 s. (*≈ 68% efficiency*)

$$E_w(p) = \frac{T(40)}{T(p)}$$

RESULTS

Assembly and FormLinearSystem, not the Krylov solve

Per-solve cost by phase (weak ladder, baseline). **Assembly and FormLinearSystem are the two longest bars**; the Krylov solve is about half of either.



CONCLUSION

Three Takeaways

89%

Strong scaling holds close to ideal

640 → 10,240 ranks (16×): 7.05 s/solve measured against an ideal 6.27 s — 89% parallel efficiency.

68%

Weak scaling reaches a solid efficiency at extreme scale

40 → 10,240 ranks, problem grows from 41.5M to 10.55B DOFs, work/rank held constant — 68% efficiency.

78%*

Assembly + FormLinearSystem, not the Krylov solve, set the floor

At 10,240 ranks (weak scaling): 70.01 s of 89.6 s instrumented per solve — 78%, and nearly flat across the ladder. Even a perfect preconditioner cannot remove this cost.

Next step: *MFEM's GPU path with partial assembly and geometric multigrid preconditioners.*

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Capped (max_it=20) Strong-Scaling Table — All Components

NP	Elast Asm	FormLS	CG Solve	Stress	Ph.Asm	Ph.FormLS	Ph.Solve	Total	Iters	Elastic-only%	Asm family%
640	24.70	21.81	11.79	3.53	21.44	0.23	1.41	84.92	20 (capped)	54.8%	80.3%
1,280	11.69	10.93	6.34	1.81	7.92	0.13	0.79	39.60	20 (capped)	57.1%	77.4%
2,560	6.18	5.61	3.31	0.98	3.38	0.06	0.42	19.94	20 (capped)	59.1%	76.3%
3,840	4.29	4.05	2.38	0.66	2.06	0.04	0.34	13.81	20 (capped)	60.3%	75.5%
5,120	3.12	3.08	2.03	0.49	1.47	0.03	0.28	10.51	20 (capped)	59.1%	73.3%
6,400	2.57	2.41	1.72	0.44	1.20	0.03	0.29	8.65	20 (capped)	57.5%	71.8%
10,240	1.56	1.64	1.44	0.26	0.71	0.02	0.21	5.85	20 (capped)	54.8%	67.3%

Seconds per solve. “Elastic-only%” = (Elast Asm + FormLS)/Total. “Asm family%” = Elast Asm + FormLS + Ph.Asm + Ph.FormLS, all over Total.

Capped (max_it=20) Weak-Scaling Table — All Components

NP	Elast Asm	FormLS	CG Solve	Stress	Ph.Asm	Ph.FormLS	Ph.Solve	Total	Iters	Elastic-only%	Asm family%
40	19.37	18.62	9.95	3.27	12.79	0.18	0.92	65.10	20 (capped)	58.4%	78.3%
80	19.43	18.55	10.70	3.36	16.55	0.17	0.98	69.74	20 (capped)	54.5%	78.4%
160	19.37	18.60	10.65	3.32	17.29	0.18	0.97	70.37	20 (capped)	54.0%	78.8%
320	21.91	20.21	10.73	3.31	17.77	0.21	1.12	75.26	20 (capped)	56.0%	79.9%
640	23.37	20.26	11.64	3.60	18.71	0.21	1.26	79.05	20 (capped)	55.2%	79.1%
1,280	23.30	20.79	11.70	3.48	20.77	0.21	1.28	81.52	20 (capped)	54.1%	79.8%
2,560	23.44	22.04	11.59	3.68	20.37	0.24	1.42	82.79	20 (capped)	54.9%	79.8%
5,120	23.65	22.84	12.71	3.75	22.82	0.23	1.69	87.70	20 (capped)	53.0%	79.3%
10,240	24.63	22.43	14.01	3.74	23.05	0.25	1.69	89.80	20 (capped)	52.4%	78.3%

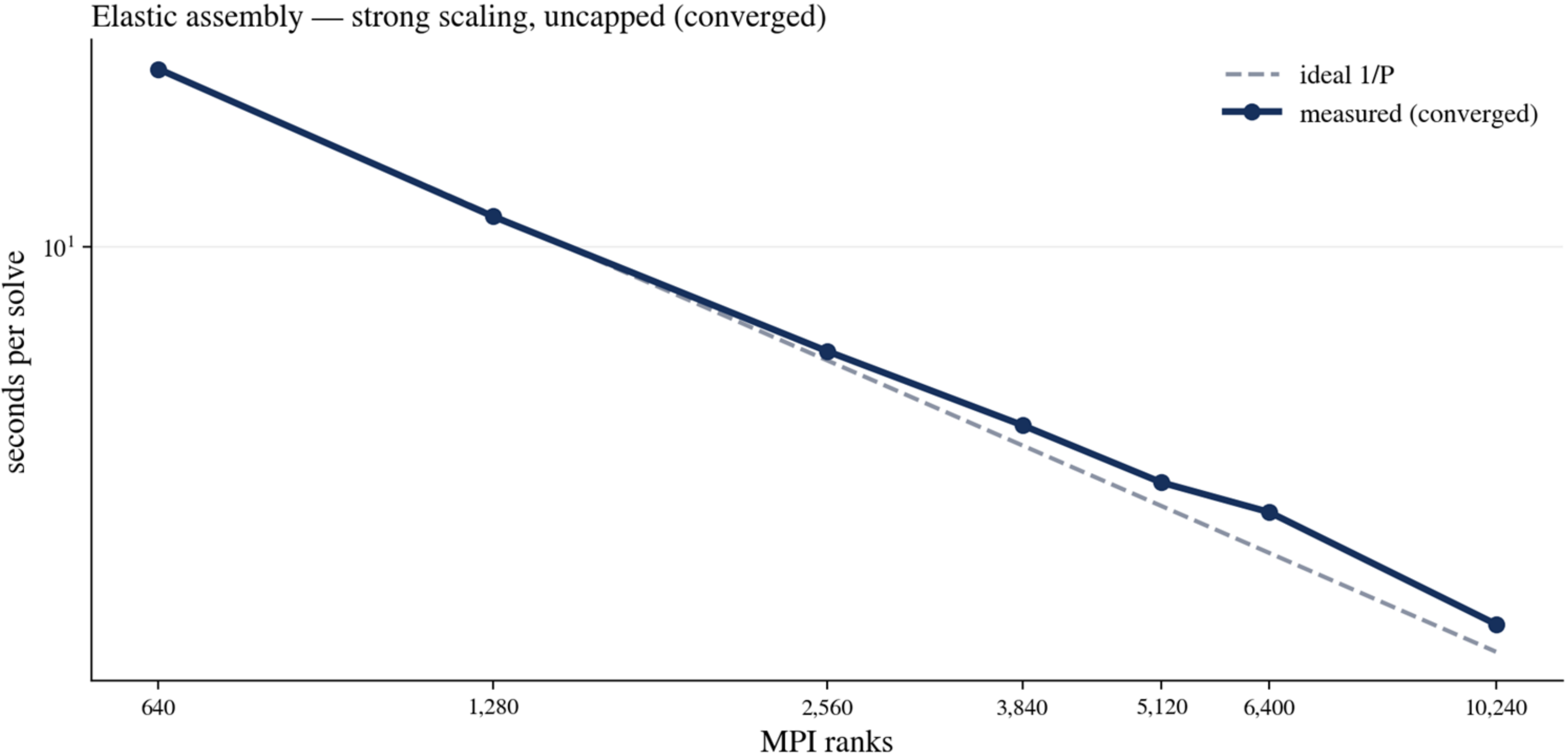
Seconds per solve. “Elastic-only%” = (Elast Asm + FormLS)/Total. “Asm family%” = Elast Asm + FormLS + Ph.Asm + Ph.FormLS, all over Total.

Uncapped Strong-Scaling Table — All Components

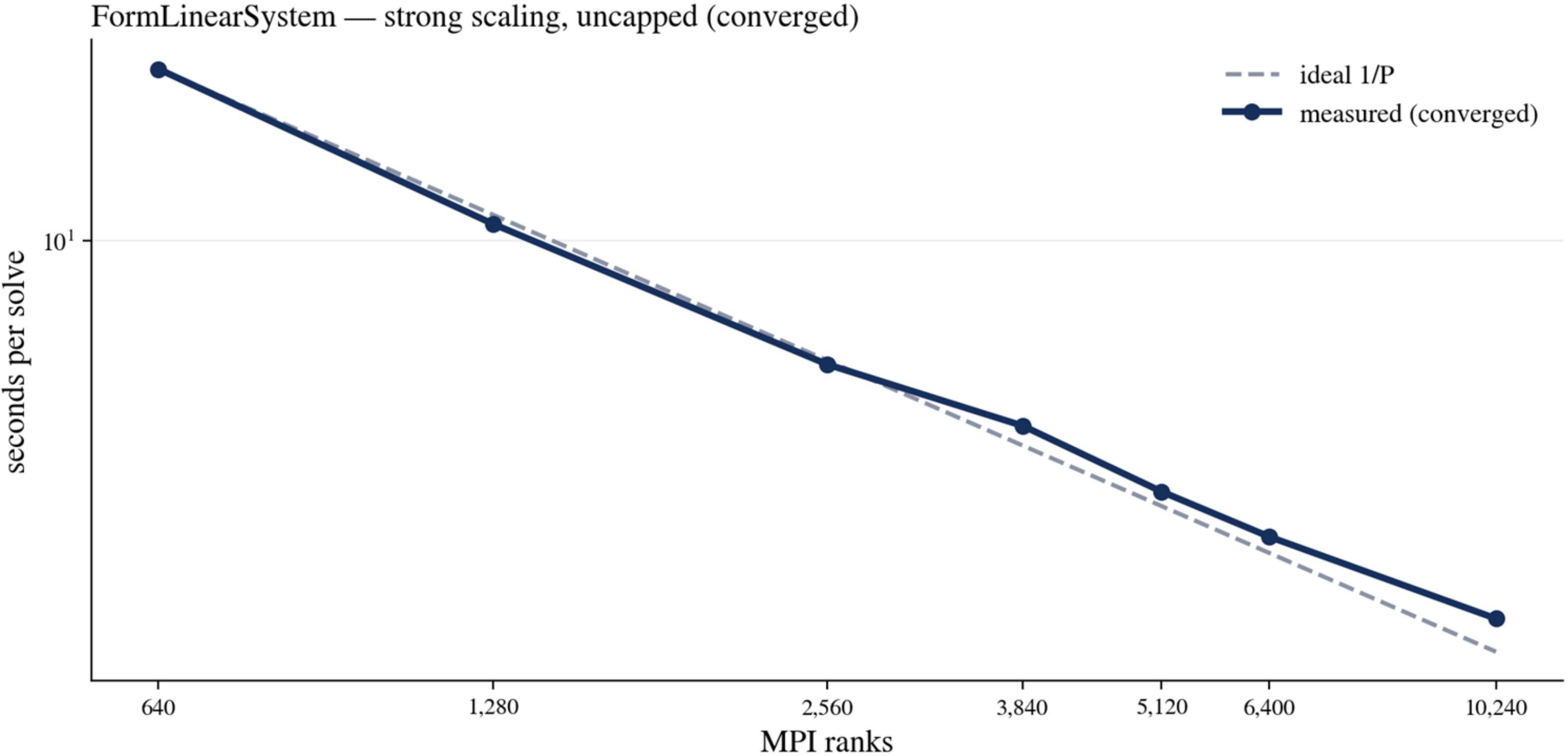
NP	Elast Asm	FormLS	CG Solve	Stress	Ph.Asm	Ph.FormLS	Ph.Solve	Total	Iters	Elastic-only%	Asm family%
640	23.28	22.65	22.05	3.58	23.74	0.23	1.44	96.97	52	47.4%	72.1%
1,280	11.55	10.84	12.06	1.95	7.84	0.13	0.80	45.17	54	49.6%	67.2%
2,560	6.08	5.56	6.74	0.98	3.51	0.06	0.42	23.35	63	49.8%	65.1%
3,840	4.27	4.15	4.90	0.66	2.19	0.04	0.35	16.56	62	50.8%	64.3%
5,120	3.26	3.03	3.62	0.50	1.57	0.03	0.28	12.29	55	51.2%	64.3%
6,400	2.83	2.45	2.96	0.40	1.19	0.03	0.29	10.14	52	52.0%	64.0%
10,240	1.66	1.66	2.28	0.26	0.74	0.02	0.19	6.80	52	48.7%	59.8%

Seconds per solve. “Elastic-only%” = (Elast Asm + FormLS)/Total. “Asm family%” = Elast Asm + FormLS + Ph.Asm + Ph.FormLS, all over Total.

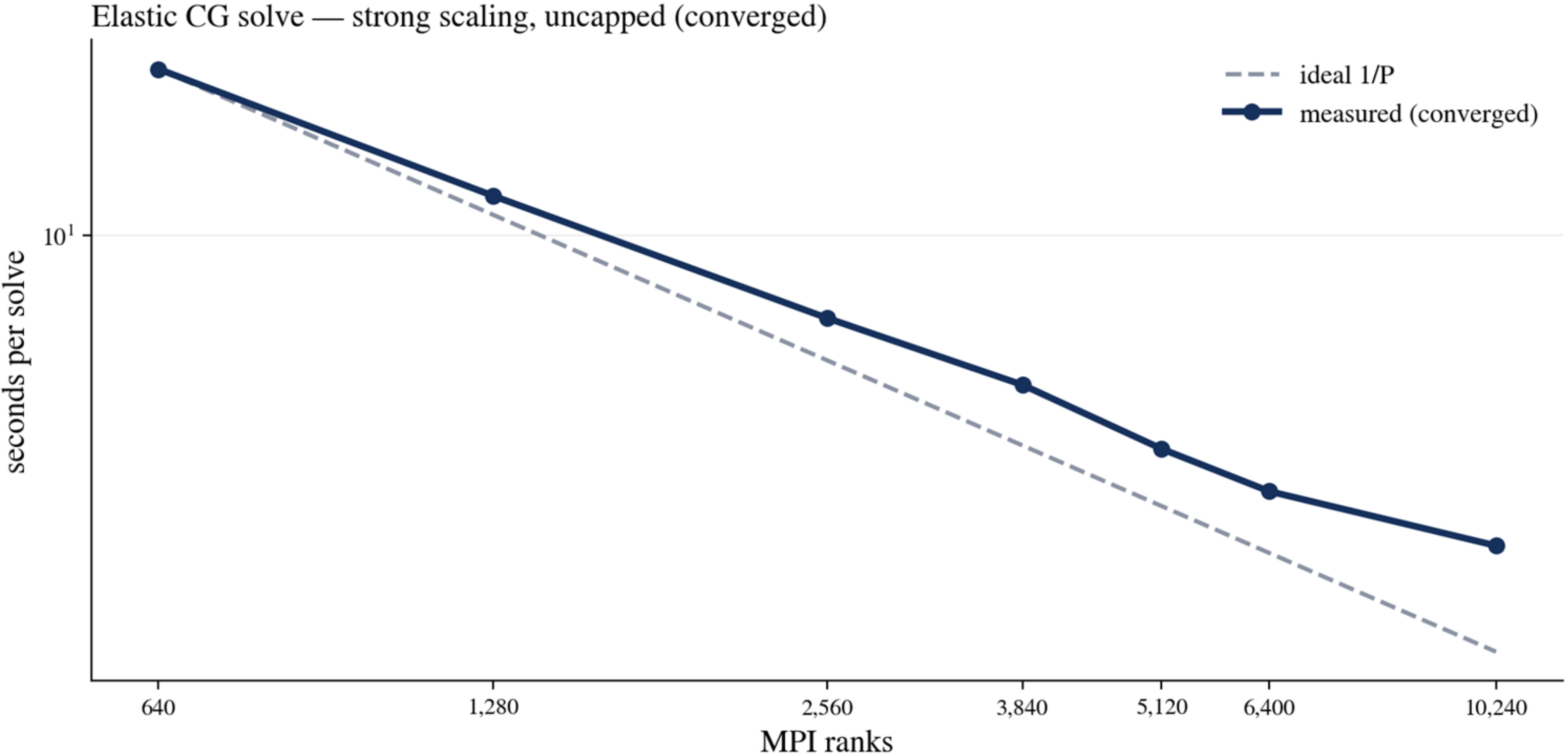
Elastic Assembly



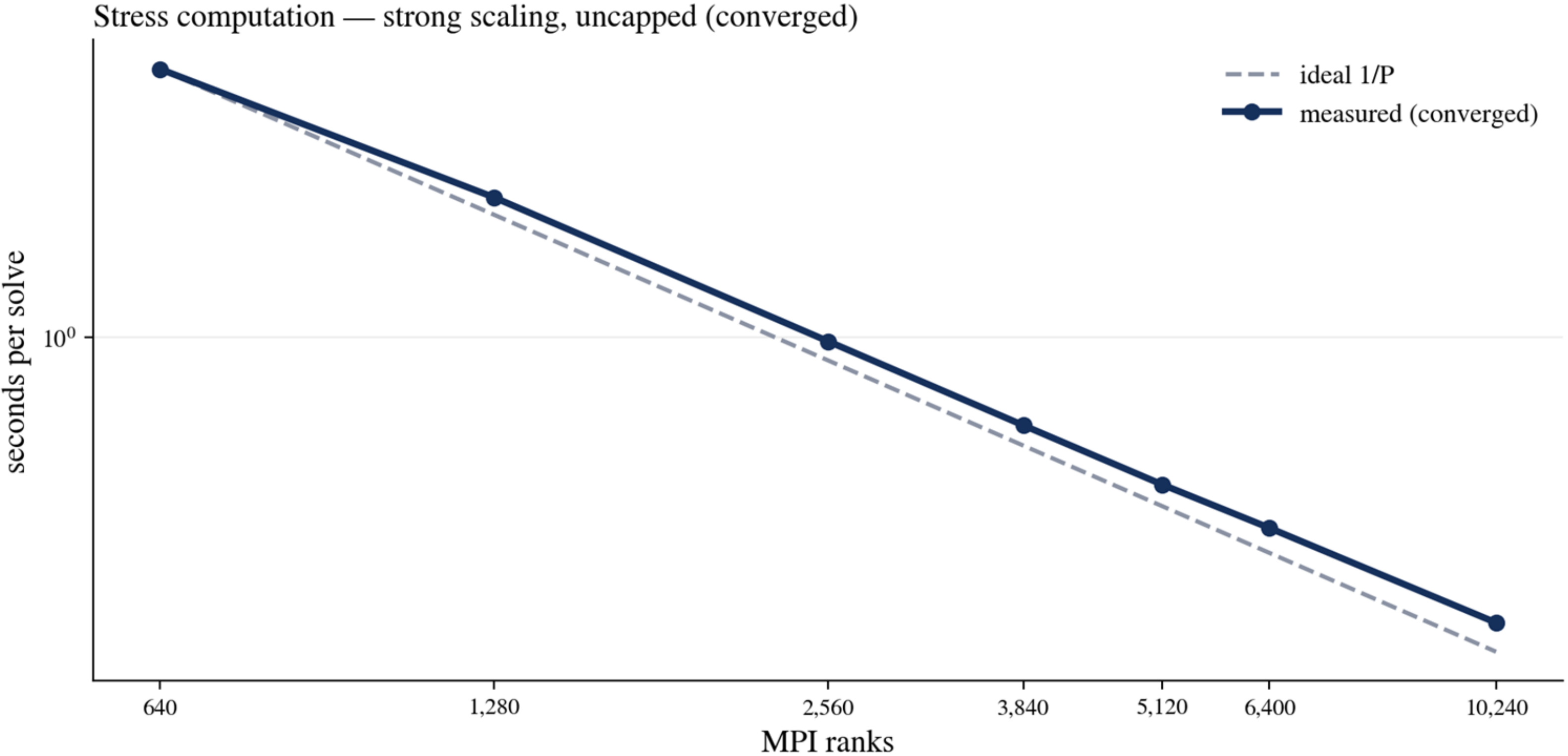
FormLinearSystem



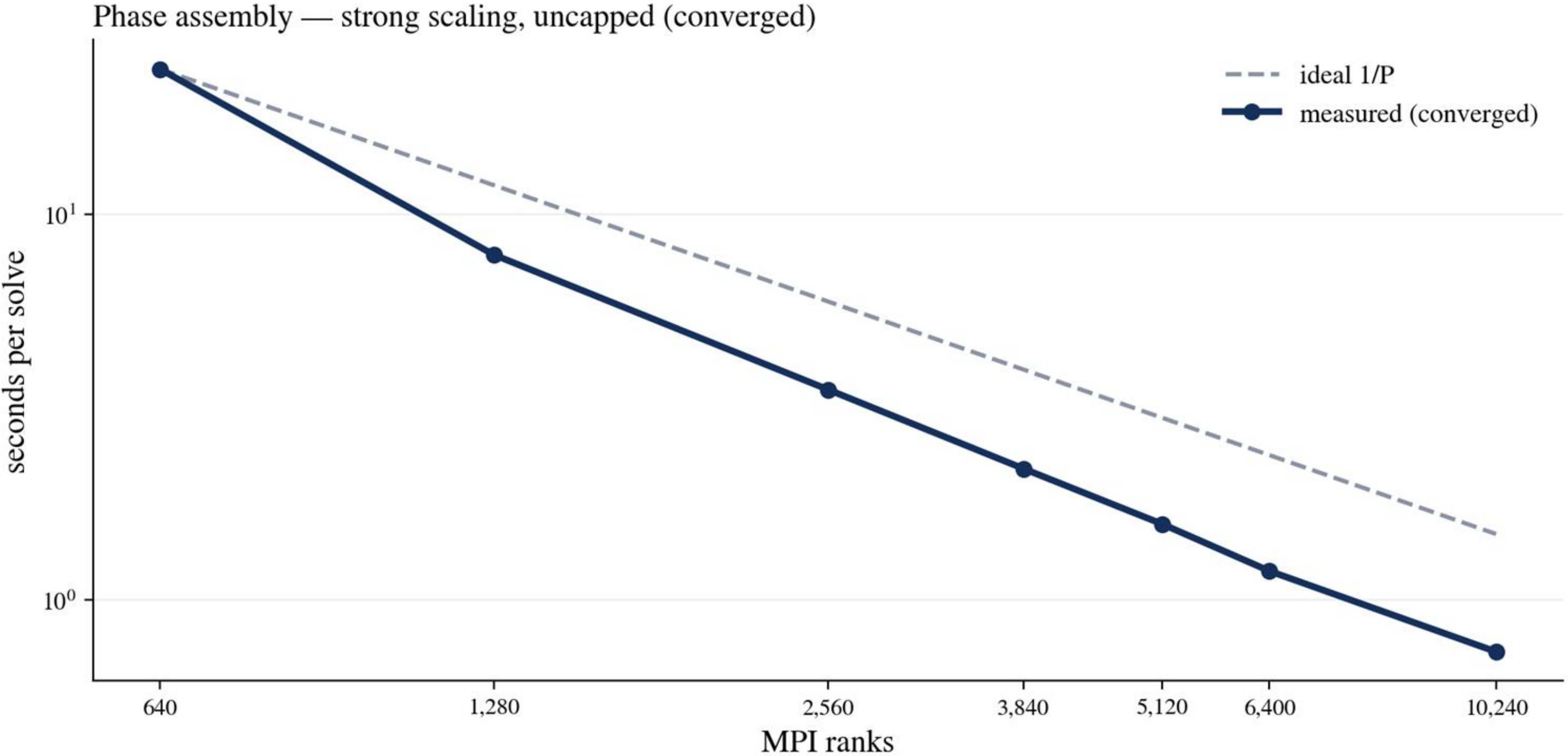
Elastic CG Solve



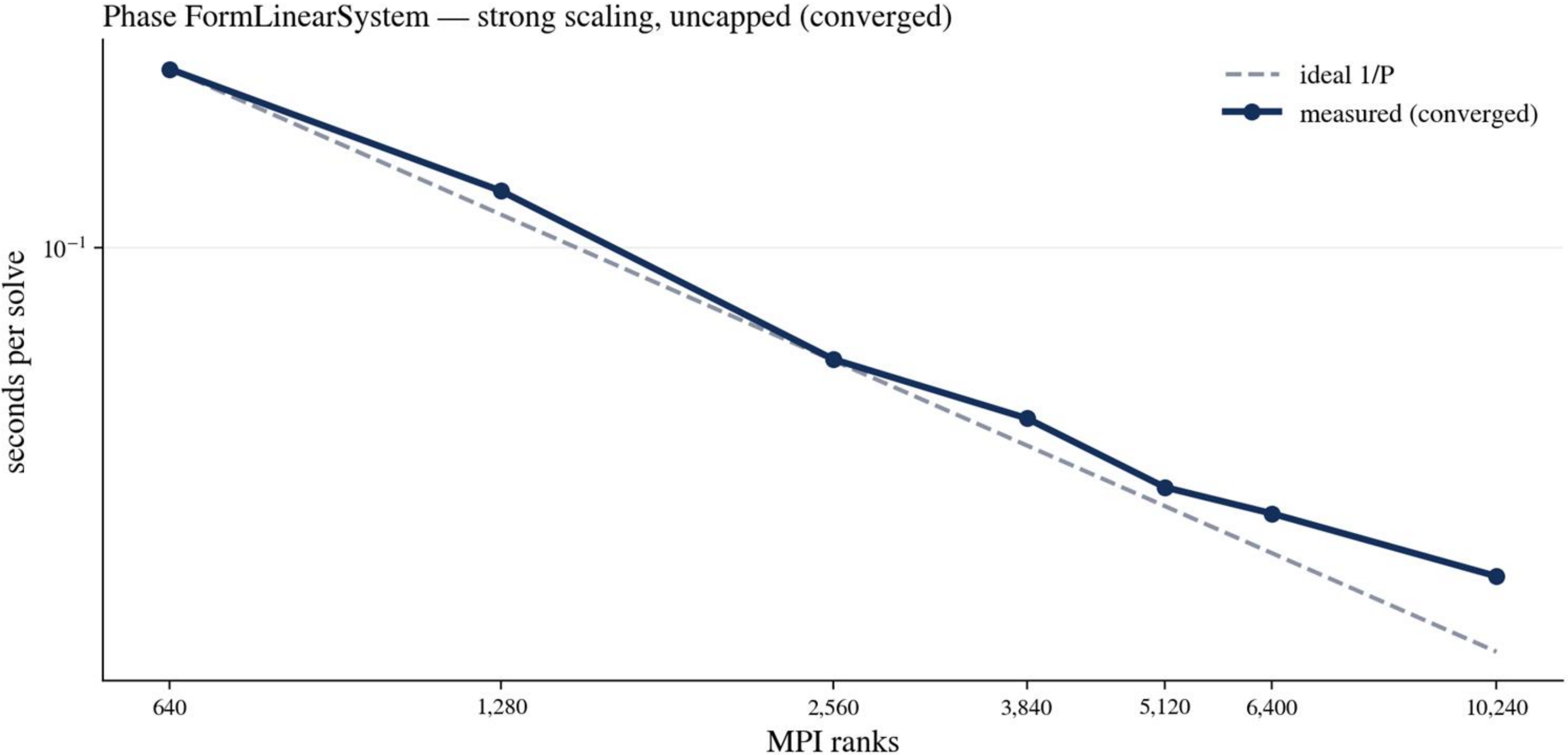
Stress Computation



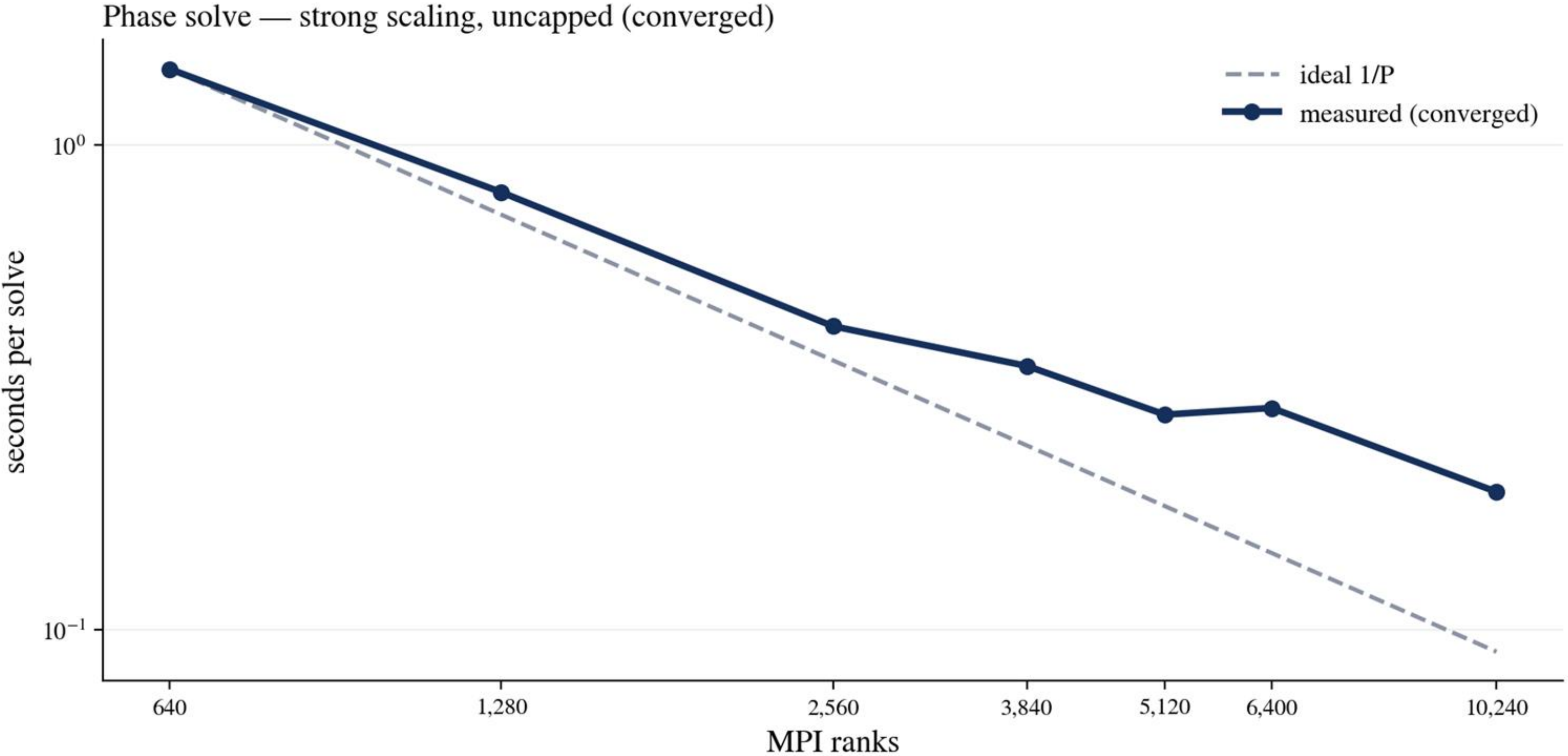
Phase Assembly



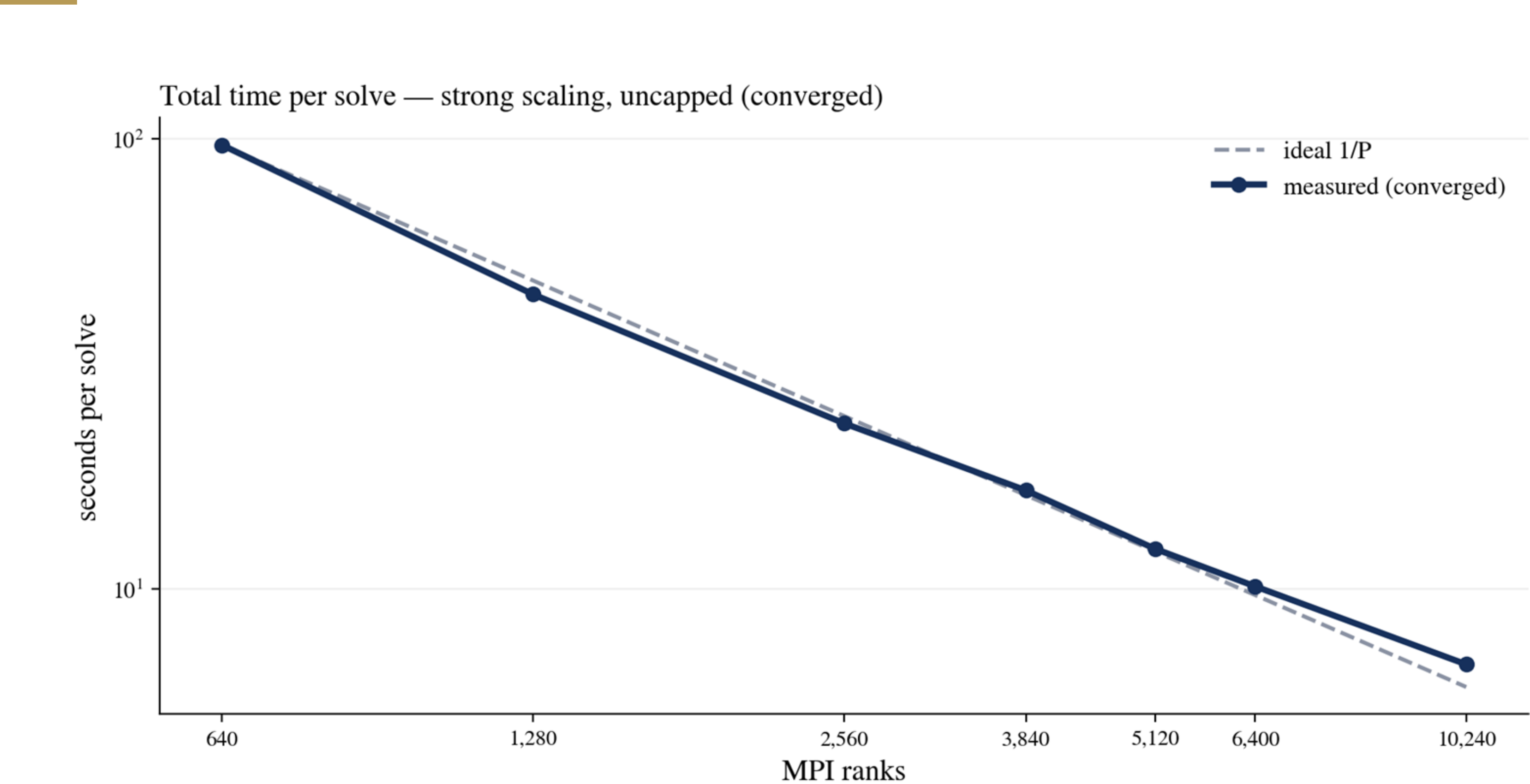
Phase FormLinearSystem



Phase Solve



Total Time Per Solve

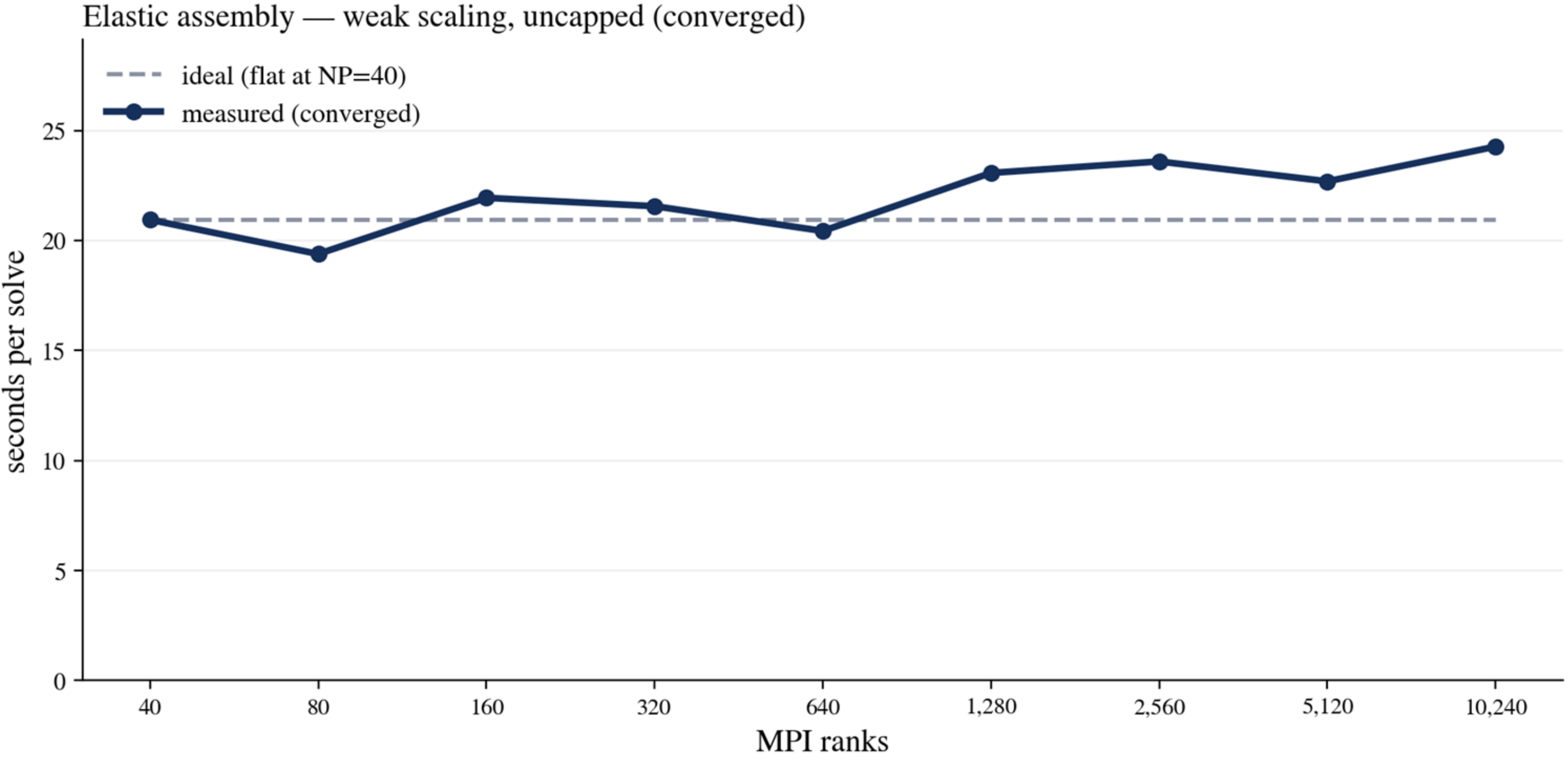


Uncapped Weak-Scaling Table — All Components

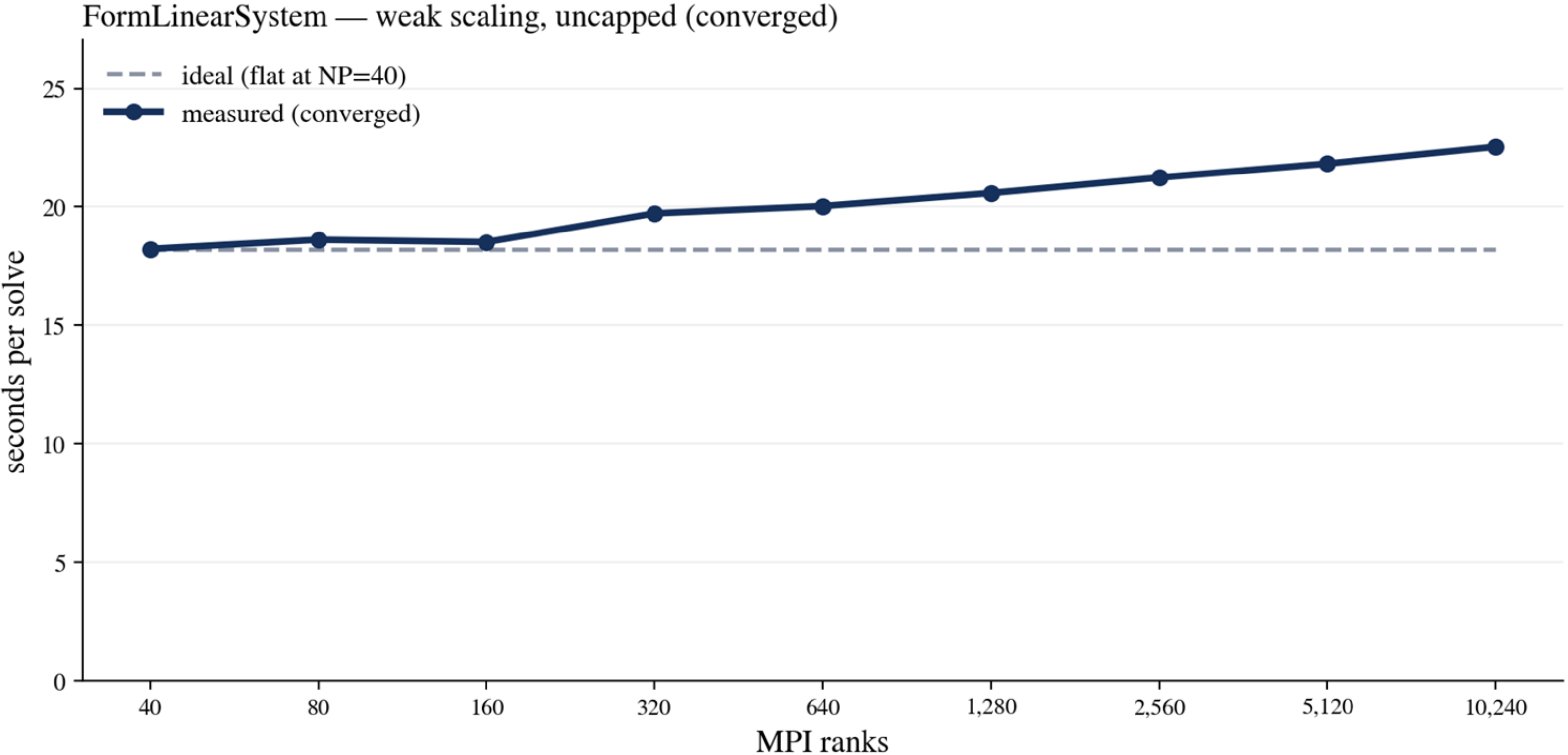
NP	Elast Asm	FormLS	CG Solve	Stress	Ph.Asm	Ph.FormLS	Ph.Solve	Total	Iters	Elastic-only%	Asm family%
40	20.94	18.22	15.73	3.29	12.89	0.18	0.91	72.16	40	54.3%	72.4%
80	19.38	18.61	17.15	3.28	16.90	0.17	0.94	76.45	44	49.7%	72.0%
160	21.93	18.51	17.29	3.29	16.97	0.18	0.95	79.13	44	51.1%	72.8%
320	21.55	19.73	19.93	3.50	19.43	0.20	1.07	85.41	50	48.3%	71.3%
640	20.43	20.03	22.73	3.80	18.67	0.21	1.23	87.10	56	46.5%	68.1%
1,280	23.07	20.58	24.51	3.40	19.98	0.22	1.31	93.07	62	46.9%	68.6%
2,560	23.59	21.24	25.78	3.54	20.02	0.25	1.48	95.88	68	46.7%	67.9%
5,120	22.68	21.82	27.10	3.86	23.15	0.23	1.63	100.48	66	44.3%	67.6%
10,240	24.26	22.54	30.64	3.86	25.70	0.25	1.65	108.91	72	43.0%	66.8%

Seconds per solve. “Elastic-only%” = (Elast Asm + FormLS)/Total. “Asm family%” = Elast Asm + FormLS + Ph.Asm + Ph.FormLS, all over Total.

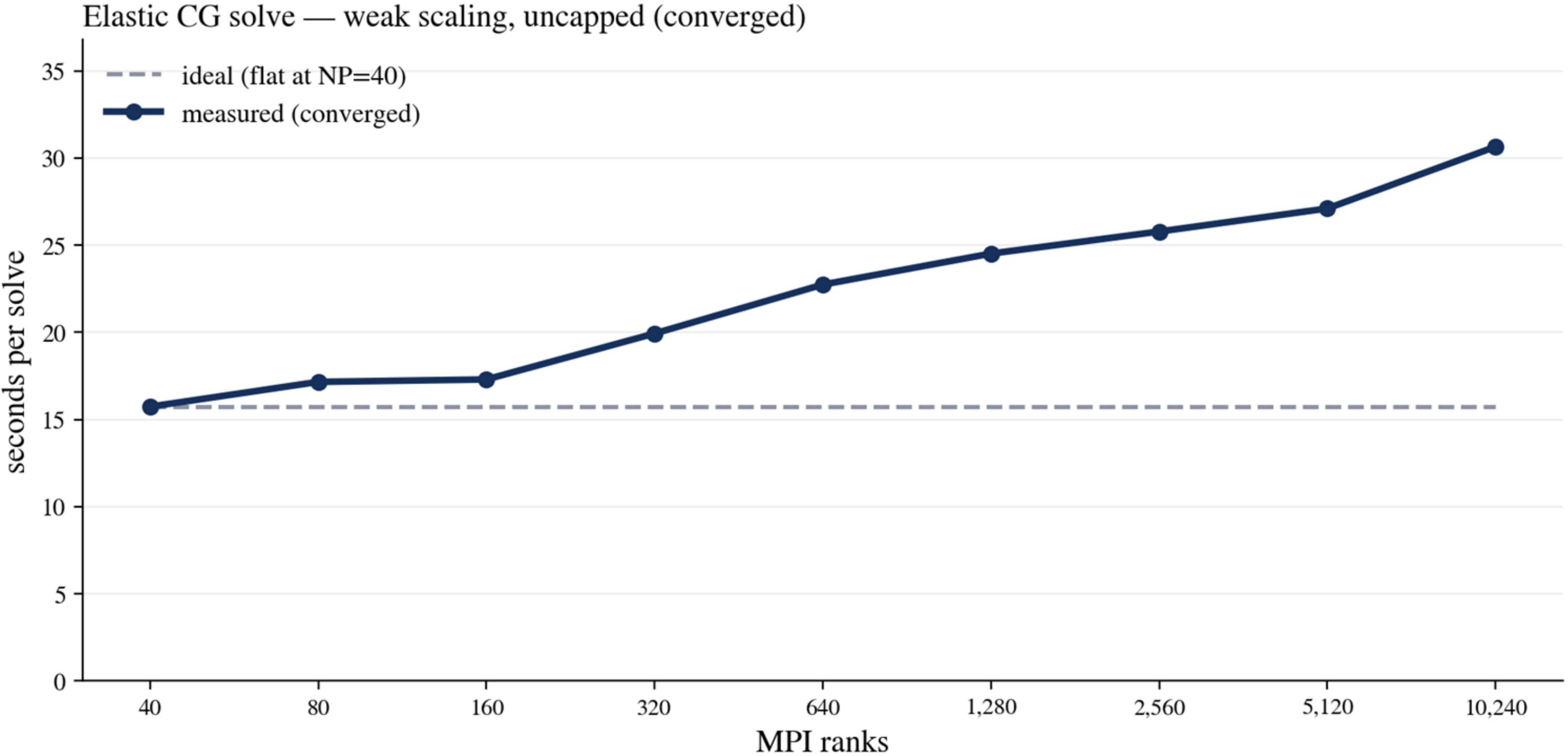
Elastic Assembly



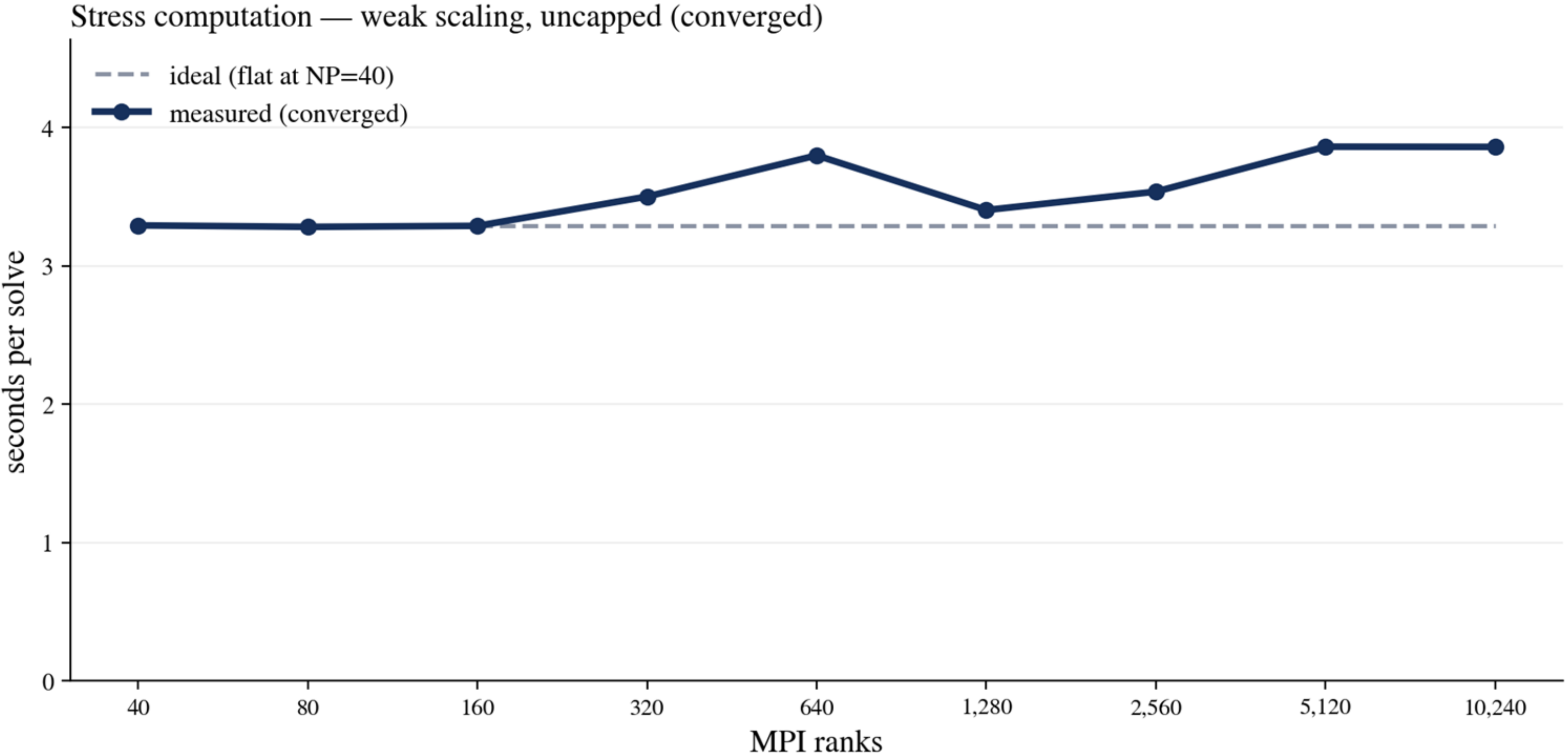
FormLinearSystem



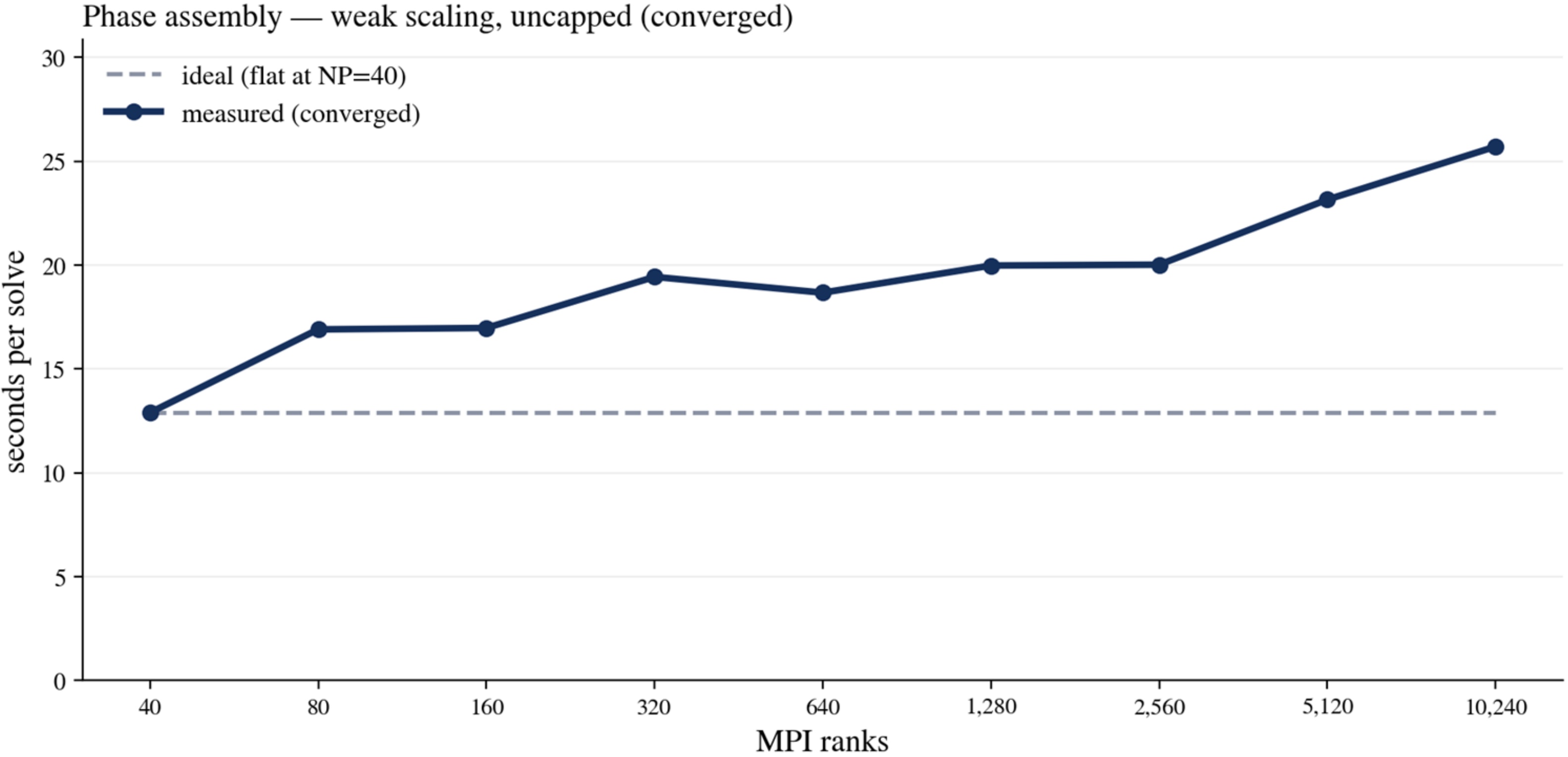
Elastic CG Solve



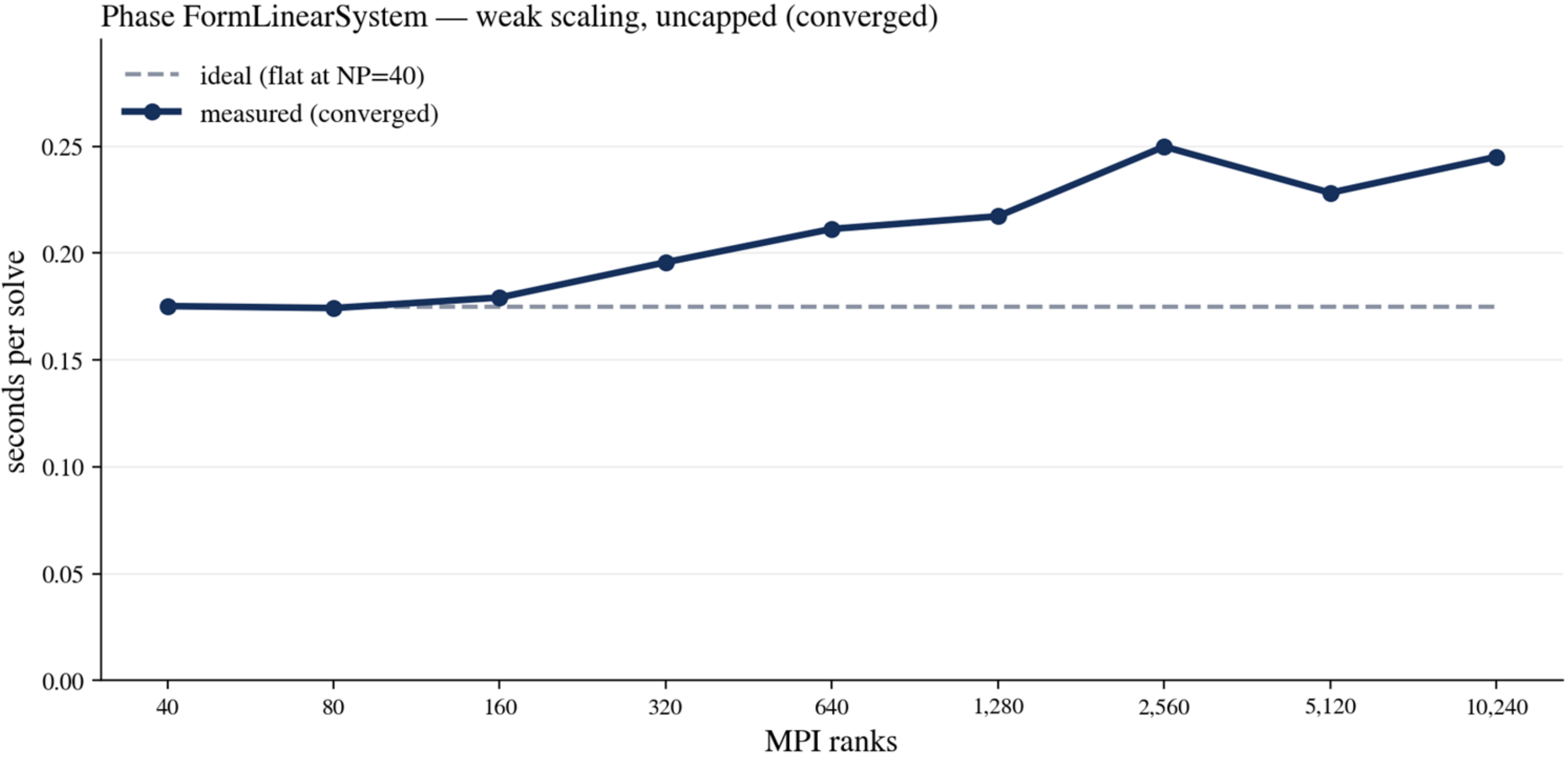
Stress Computation



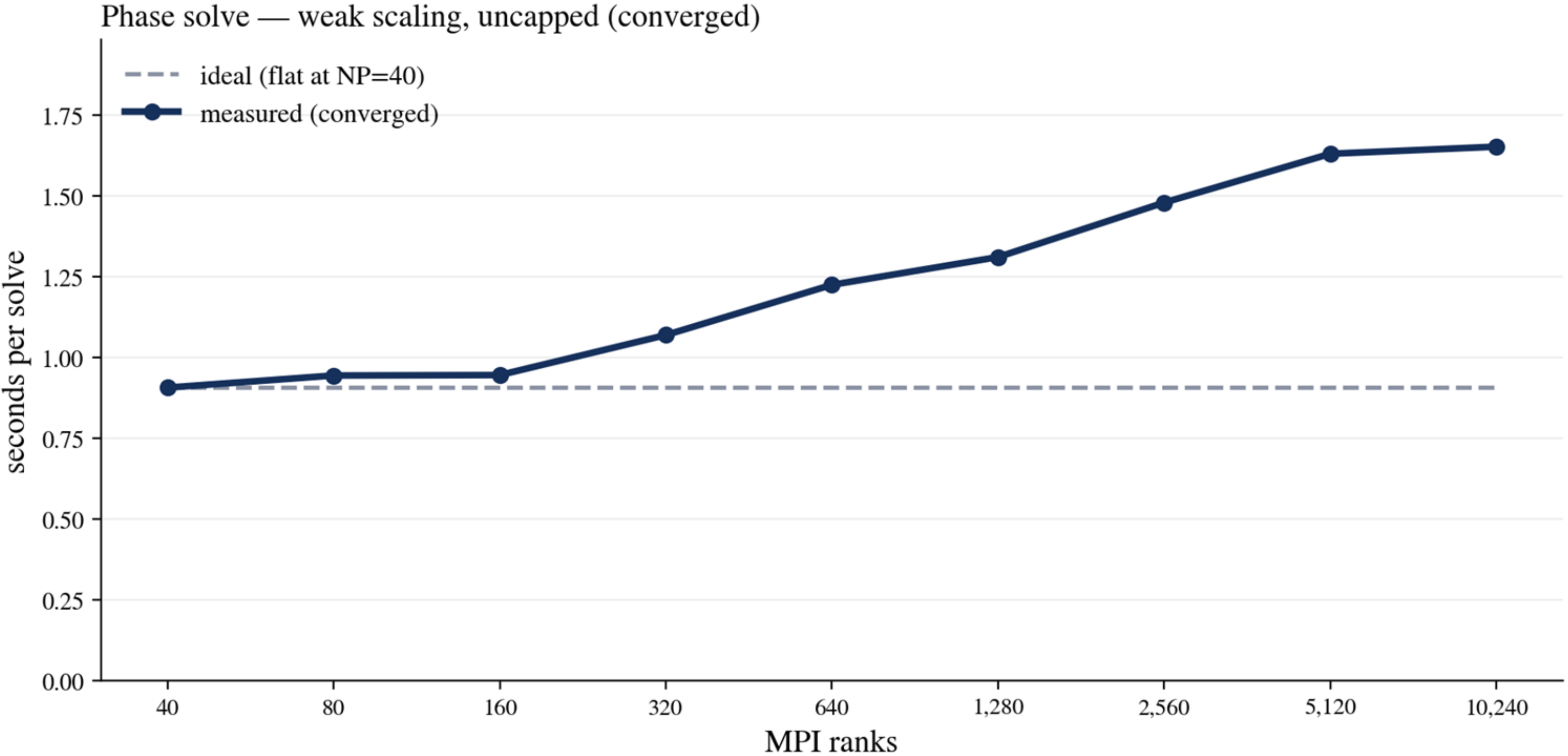
Phase Assembly



Phase FormLinearSystem



Phase Solve



Total Time Per Solve

